1. A (1, –1, –3), B (2, 1, –2) and C (–5, 2, –6) are the position vectors of the vertices of a triangle ABC. The length of the bisector of its internal angle at A is :

(A) $\sqrt{10}/4$ (B) $3\sqrt{10}/4$ (C) $\sqrt{10}$ (D) None

यदि A (1, –1, –3), B (2, 1, –2) तथा C (–5, 2, –6) एक त्रिभुज ABC के शीर्षो के स्थिति सदिश है। कोण A के आन्तरिक कोणार्द्धक की लम्बाई होगी :

(A) $\sqrt{10}/4$ (B) $3\sqrt{10}/4$ (C) $\sqrt{10}$ (D) कोई नहीं

Ans. B

Sol. $\overrightarrow{AB} = (1,2,1)$ $\Rightarrow \left|\overrightarrow{AB}\right| = \sqrt{6}$

$\overrightarrow{AC} = (-6,3,-3)$ $\Rightarrow \left|\overrightarrow{AC}\right| = 3\sqrt{6}$

$\frac{BD}{DC} = \frac{AB}{AC} = \frac{1}{3} \rightarrow$ Angle Bisector

$\overrightarrow{AD} = \frac{3\overrightarrow{AB} + \overrightarrow{AC}}{4} = \frac{1}{4}(-3,9,0)$

$\left|\overrightarrow{AD}\right| = \frac{3}{4}\sqrt{10}$

2. Let $\vec{p}$ is the p.v. of the orthocentre and $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is the origin. If $\vec{p} = K\vec{g}$, then K =

त्रिभुज ABC में माना लम्बकेन्द्र का स्थिति सदिश $\vec{p}$ है तथा केन्द्रक का स्थिति सदिश $\vec{g}$ है जहाँ परिकेन्द्र मूल बिन्दु है। यदि $\vec{p} = K\vec{g}$, तो K =

(A) 3 (B) 2 (C) 1/3 (D) 2/3

Ans. A

Sol. $\Rightarrow \vec{g} = \frac{\vec{p} + \vec{o}}{3}$

$\Rightarrow \vec{p} = 3\vec{g}$

3. A vector $\vec{a}$ has components 2p and 1 with respect to a rectangular cartesian system. The system is rotated through

a certain angle about the origin in the counterclockwise sense. If with respect to the new system, $\vec{a}$ has components $p+1$ and 1 then :

(A) $p = 0$ (B) $p = 1$ or $p = -1/3$ (C) $p = -1$ or $p = 1/3$ (D) $p = 1$ or $p = -1$

आयताकार कार्तिकेय निकाय में सदिश $\vec{a}$ के घटक 2p तथा 1 है। निकाय को वामावर्त दिशा में मूल बिन्दु के सापेक्ष किसी कोण से घूमा दिया जाता है। यदि नये निकाय में, $\vec{a}$ के घटक $p+1$ तथा 1 तो :

(A) $p = 0$ (B) $p = 1$ or $p = -1/3$ (C) $p = -1$ or $p = 1/3$ (D) $p = 1$ or $p = -1$

Ans. B

Sol. Equate the magnitude i.e. $4p^2 + 1 = (p + 1)^2 + 1 = p^2 + 2p + 2$

$\Rightarrow \quad 3p^2 - 2p - 1 = 0 \Rightarrow p = 1$ or $-1/3$

4. Four points A(+1, –1, 1) ; B(1, 3, 1) ; C(4, 3, 1) and D(4, – 1, 1) taken in order are the vertices of :

(A) a parallelogram which is neither a rectangle nor a rhombus

(B) rhombus

(C) an isosceles trapezium

(D) a cyclic quadrilateral

चार बिन्दु A(+1, –1, 1) ; B(1, 3, 1) ; C(4, 3, 1) तथा D(4, – 1, 1) क्रमवार शीर्ष है :

(A) एक सामान्तर चतुर्भुज के जो ना तो आयत है न ही समचतुर्भुज है

(B) समचतुर्भुज

(C) एक समद्विबाहु समलम्ब चतुर्भुज

(D) एक चक्रिय चतुर्भुज

Ans. D

Sol. $4P^2 + 1 = (P + 1)^2 + 1 \Rightarrow 4P^2 - (P + 1)^2 = 0$

$\Rightarrow (3P + 1)(P - 1)$

$\Rightarrow P = \dfrac{-1}{3}$ or $P = 1$

5. Let α, β and γ be distinct real numbers. The points whose position vector's are $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$; $\beta\hat{i} + \gamma\hat{j} + \alpha\hat{k}$ and $\gamma\hat{i} + \alpha\hat{j} + \beta\hat{k}$:

(A) are collinear (B) form an equilateral triangle

(C) form a scalene triangle (D) form a right angled triangle

माना α, β तथा γ वास्तविक भिन्न संख्यायें है। बिन्दु जिनके स्थिति सदिश $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$; $\beta\hat{i} + \gamma\hat{j} + \alpha\hat{k}$ तथा $\gamma\hat{i} + \alpha\hat{j} + \beta\hat{k}$ है:

(A) समरेखीय (B) एक समबाहु त्रिभुज बनाते है

(C) विषमबाहु त्रिभुज बनाते है (D) एक समकोणीय त्रिभुज बनाते है

Ans. B

Sol. Let P (α, β, γ)
Q (β, γ, α)
C (γ, α, β)
PQ = QC = PC

6. If the vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$ constitute the sides of a ΔABC, then the length of the median bisecting the vector $\vec{c}$ is :

यदि सदिश $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ तथा $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$, त्रिभुज ABC की तीन भुजा बनाते है, तो सदिश $\vec{c}$ को समद्विभाजित करने वाली मध्यिका की लम्बाई है :

(A) $\sqrt{2}$ (B) $\sqrt{14}$ (C) $\sqrt{74}$ (D) $\sqrt{6}$

Ans. D

Sol. $\vec{a} = (3,1,-2)$

$\vec{b} = (-1,3,4)$

$\vec{c} = (4,-2,-6)$

$\vec{b} + \vec{c} = \vec{a}$
Let a (0, 0, 0)
B (4, –2, –6)
C (1, –3, –4)
mid-point of AB $\Rightarrow$ M(2, –1, –3)
$\overrightarrow{CM}(1,+2,1)$

$\left|\overrightarrow{CM}\right| = \sqrt{6}$

7. If the three points with position vectors (1, a, b) ; (a, 2, b) and (a, b, 3) are collinear in space, then the value of a + b is :

(A) 3 (B) 4 (C) 5 (D) None

यदि तीन बिन्दु जिनके स्थिति सदिश (1, a, b) ; (a, 2, b) तथा (a, b, 3) त्रिविम में संरेखीय है, तो a + b का मान है :

(A) 3 (B) 4 (C) 5 (D) कोई नहीं

Ans. B

Sol. A (1, a, b) ; B (a, 2, b) ; C (a, b, 3)

$\overrightarrow{AB} = (a-1)\,\hat{i} + (2-a)\,\hat{j} + 0\,\hat{k}$; $\overrightarrow{BC} = 0\hat{i} + (b-2)\hat{j} + (3-b)\hat{k}$

$\overrightarrow{AB} = \lambda\ \overrightarrow{BC} = \lambda\left(0\hat{i} + (b-2)\hat{j} + (3-b)\hat{k}\right)$ where $\lambda \neq 0$

hence $a - 1 = 0 \quad \Rightarrow \quad a = 1$ _____(1)

$2 - a = \lambda(b-2)$ _____(2)

and $3 - b = 0 \quad \Rightarrow \quad b = 3$ _____(3)

with $a = 1$ and $b = 3$, $\lambda = 1$

hence $a + b = 4 \quad \Rightarrow$ (B)

8. Consider the following 3 lines in space

$L_1 : \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k} + \lambda(2\hat{i} + 4\hat{j} - \hat{k})$

$L_2 : \vec{r} = \hat{i} + \hat{j} - 3\hat{k} + \mu(4\hat{i} + 2\hat{j} + 4\hat{k})$

$L_3 : \vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k} + t(2\hat{i} + \hat{j} + 2\hat{k})$

Then which one of the following pair(s) are in the same plane :

(A) only L_1L_2 (B) only L_2L_3 (C) only L_3L_1 (D) L_1L_2 and L_2L_3

माना 3 रेखाऐं त्रिविम है

$L_1 : \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k} + \lambda(2\hat{i} + 4\hat{j} - \hat{k})$

$L_2 : \vec{r} = \hat{i} + \hat{j} - 3\hat{k} + \mu(4\hat{i} + 2\hat{j} + 4\hat{k})$

$L_3 : \vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k} + t(2\hat{i} + \hat{j} + 2\hat{k})$

तो निम्न में से कौनसा युग्म समतलीय है :

(A) केवल L_1L_2 (B) केवल L_2L_3 (C) केवल L_3L_1 (D) L_1L_2 तथा L_2L_3

Ans. D

Sol. L_1L_2 intersecting ; L_2L_3 parallel ; L_3L_1 skew

9. The vectors $3\hat{i} - 2\hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} + 5\hat{k}$ and $2\hat{i} + \hat{j} - 4\hat{k}$ form the sides of a triangle. Then triangle is :

(A) an acute angled triangle (B) an obtuse angled triangle

(C) an equilateral triangle (D) a right angled triangle

सदिश $3\hat{i} - 2\hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} + 5\hat{k}$ तथा $2\hat{i} + \hat{j} - 4\hat{k}$ त्रिभुज की भुजाऐं बनाते है, तो त्रिभुज है :

(A) न्यूकोणीय (B) अधिक कोणीय

(C) समबाहु (D) समकोणीय

Ans. D

Sol. $\vec{a} = (3,-2,1) \Rightarrow |\vec{a}| = \sqrt{14}$

$\vec{b} = (1,-3,5) \Rightarrow |\vec{b}| = \sqrt{35}$

$\vec{c} = (2,1,-4) \Rightarrow |\vec{c}| = \sqrt{21}$

$\vec{a} \cdot \vec{c} = 0$

10. If A(0, 1, 0), B (0, 0, 0), C(1, 0, 1) are the vertices of a ΔABC. Match the entries of column-I with column-II :

Column-I	**Column-II**
(A) Orthocentre of ΔABC	(P) $\frac{\sqrt{2}}{2}$
(B) Circumcentre of ΔABC	(Q) $\frac{\sqrt{3}}{2}$
(C) Area (ΔABC)	(R) $\frac{\sqrt{3}}{3}$
(D) Distance between orthocentre and centroid	(S) $\frac{\sqrt{3}}{6}$
(E) Distance between orthocentre and circumcentre	(T) (0, 0, 0)
(F) Distance between circumcentre and centroid	(U) $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$
(G) Incentre of ΔABC	(V) $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
(H) Centroid of ΔABC	(W) $\left(\frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}\right)$

यदि A(0, 1, 0), B (0, 0, 0), C(1, 0, 1) त्रिभुज ABC के शीर्ष है। Column-I से Column-II का मिलान कीजिए :

Column-I	**Column-II**
(A) ΔABC का लंबकेन्द्र	(P) $\frac{\sqrt{2}}{2}$
(B) ΔABC का परिकेन्द्र	(Q) $\frac{\sqrt{3}}{2}$
(C) क्षेत्रफल (ΔABC)	(R) $\frac{\sqrt{3}}{3}$

(D) लम्बकेन्द्र तथा केन्द्रक के मध्य दूरी (S) $\frac{\sqrt{3}}{6}$

(E) लम्बकेन्द्र तथा परिकेन्द्र के मध्य दूरी (T) (0, 0, 0)

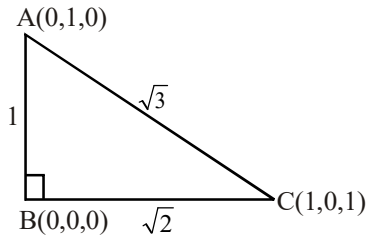
(F) परिकेन्द्र तथा केन्द्रक के मध्य दूरी (U) $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

(G) ΔABC का अन्तःकेन्द्र (V) $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

(H) ΔABC का केन्द्रक (W) $\left(\frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}\right)$

Ans. (A) T; (B) U; (C) P; (D) R; (E) Q; (F) S; (G) W; (H) V

Sol.



Clearly ΔABC is right angled at B.

(A) Orthocentre = (0, 0, 0)

(B) B (0,0,0) (Orthocentre) — 2 : 1 — G(centroid) — D($\vec{d}$) (Circumcentre)

We have $\frac{2\vec{d}}{3} = \frac{1}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{1}{3}\hat{k}$

$\Rightarrow$ Circumcentre = $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

(C) Area (ΔABC) = $\frac{1}{2}$ (AB) (BC) = $\frac{1}{2}$ (1) $(\sqrt{2})$ = $\frac{\sqrt{2}}{2}$ sq. unit

(D) Distance between orthocentre and centroid

$= \sqrt{\left(\frac{1}{3}-0\right)^2 + \left(\frac{1}{3}-0\right)^2 + \left(\frac{1}{3}-0\right)^2} = \sqrt{\frac{1}{9}+\frac{1}{9}+\frac{1}{9}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$ unit.

(E) Distance between orthocentre and circumcentre

$$= \sqrt{\left(\frac{1}{2}-0\right)^2+\left(\frac{1}{2}-0\right)^2+\left(\frac{1}{2}-0\right)^2} = \sqrt{\frac{1}{4}+\frac{1}{4}+\frac{1}{4}} = \frac{\sqrt{3}}{2} \text{ unit.}$$

(F) Distance between centroid and circumcentre

$$= \sqrt{\left(\frac{1}{2}-\frac{1}{3}\right)^2+\left(\frac{1}{2}-\frac{1}{3}\right)^2+\left(\frac{1}{2}-\frac{1}{3}\right)^2} = \sqrt{\frac{1}{36}+\frac{1}{36}+\frac{1}{36}} = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6} \text{ unit.}$$

(G) We know that position vector of incentre of $\Delta ABC = \dfrac{a\vec{a}+b\vec{b}+c\vec{c}}{a+b+c}$

$\therefore$ Position vector of incentre $= \dfrac{\sqrt{2}(\hat{j})+\sqrt{3}(\vec{0})+1(\hat{i}+\hat{k})}{\sqrt{1}+\sqrt{2}+\sqrt{3}} = \dfrac{\hat{i}+\sqrt{2}\,\hat{j}+\hat{k}}{\sqrt{1}+\sqrt{2}+\sqrt{3}}$

$\therefore$ Incentre $= \left(\dfrac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \dfrac{\sqrt{2}}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \dfrac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}\right)$

(H) centriod $= \left(\dfrac{1}{3}, \dfrac{1}{3}, \dfrac{1}{3}\right)$

1. The acute angle between the medians drawn from the acute angles of an isosceles right angled triangle is :

(A) $\cos^{-1}\left(\frac{2}{3}\right)$ (B) $\cos^{-1}\left(\frac{3}{4}\right)$ (C) $\cos^{-1}\left(\frac{4}{5}\right)$ (D) None

एक समद्विबाहु समकोणीय त्रिभुज के न्यूनकोणो से मध्यिकाओं के मध्य न्यूनकोण है :

(A) $\cos^{-1}\left(\frac{2}{3}\right)$ (B) $\cos^{-1}\left(\frac{3}{4}\right)$ (C) $\cos^{-1}\left(\frac{4}{5}\right)$ (D) कोई नहीं

Ans. C

Sol. $\overrightarrow{OA}=(0,a)$, $\overrightarrow{OB}=(a,0)$

$\overrightarrow{BC}=\left(-a,\frac{a}{2}\right)$, $\overrightarrow{AD}=\left(\frac{a}{2},-a\right)$

$$\cos\theta=\frac{\overrightarrow{AD}\cdot\overrightarrow{BC}}{\left|\overrightarrow{AD}\right|\left|\overrightarrow{BC}\right|}=\frac{\frac{a^2}{2}+\frac{a^2}{2}}{\frac{5}{4}a^2}=\frac{4}{5}$$

$$\theta=\cos^{-1}\left(\frac{4}{5}\right)$$

2. If the vectors $3\bar{p}+\bar{q}\,;5\bar{p}-3\bar{q}$ and $2\bar{p}+\bar{q}\,;4\bar{p}-2\bar{q}$ are pairs of mutually perpendicular vectors then $\sin\left(\bar{p}\,\hat{}\,\bar{q}\right)$ is:

यदि सदिश $3\bar{p}+\bar{q}\,;5\bar{p}-3\bar{q}$ तथा $2\bar{p}+\bar{q}\,;4\bar{p}-2\bar{q}$ युग्म में आपस में लम्बवत् है तथा $\sin\left(\bar{p}\,\hat{}\,\bar{q}\right)$ है :

(A) $\frac{\sqrt{155}}{4}$ (B) $\frac{\sqrt{55}}{8}$ (C) $\frac{3}{16}$ (D) $\frac{\sqrt{247}}{16}$

Ans. B

Sol. $(3\vec{p}+\vec{q})\cdot(5\vec{p}-3\vec{q})=0$ or $15\vec{p}^{\,2}-3\vec{q}^{\,2}=4\vec{p}\cdot\vec{q}$ ....(1)

$(2\vec{p}+\vec{q})\cdot(4\vec{p}-2\vec{q})=0$ or $8\vec{p}^{\,2}=2\vec{q}^{\,2}$ $\Rightarrow$ $\vec{q}^{\,2}=4\vec{p}^{\,2}$ ....(2)

now $\cos\theta=\frac{\vec{p}\cdot\vec{q}}{|\vec{p}||\vec{q}|}$; substituting $\vec{q}^{\,2}=4\vec{p}^{\,2}$ in (1) $3\vec{p}^{\,2}=4\vec{p}\cdot\vec{q}$

$$\cos\theta = \frac{3}{4}\cdot\frac{\vec{p}^2}{|\vec{p}|2|\vec{p}|} = \frac{3}{8} \quad \Rightarrow \quad \sin\theta = \frac{\sqrt{55}}{8} \quad \Rightarrow \quad \text{(B)}$$

3. Consider the points A, B and C with position vectors $\left(-2\hat{i}+3\hat{j}+5\hat{k}\right), \left(\hat{i}+2\hat{j}+3\hat{k}\right)$ and $7\hat{i}-\hat{k}$ respectively.

Statement-1: The vector sum, $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\vec{0}$

because

Statement-2: A, B and C form the vertices of a triangle.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true

माना बिन्दु A, B तथा C जिनके स्थिति सदिश $\left(-2\hat{i}+3\hat{j}+5\hat{k}\right), \left(\hat{i}+2\hat{j}+3\hat{k}\right)$ तथा $7\hat{i}-\hat{k}$ क्रमशः है

कथन-1: सदिश योग $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\vec{0}$

क्योंकि

कथन-2: A, B तथा C त्रिभुज के शीर्ष है।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है

(C) कथन–1 सत्य है, कथन–2 असत्य है

(D) कथन–1 असत्य है, कथन– 2 सत्य है

Ans. C

Sol. $\overrightarrow{AB} = (3, -1, -2)$;

$\overrightarrow{BC} = (6, -2, -4)$

$\overrightarrow{CA} = (-9, 3, 6)$

$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA} = \vec{0}$

$\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CA} \Rightarrow$ Collinear

4. The set of values of c for which the angle between the vectors $cx\hat{i} - 6\hat{j} + 3\hat{k}$ and $x\hat{i} - 2\hat{j} + 2cx\hat{k}$ is acute for every $x \in R$ is :

c के मानों का समुच्चय जिनके लिये सदिशों $cx\hat{i} - 6\hat{j} + 3\hat{k}$ तथा $x\hat{i} - 2\hat{j} + 2cx\hat{k}$ के मध्य कोण, $x \in R$, के लिये न्यून है :

(A) (0, 4/3) (B) [0, 4/3] (C) (11/9, 4/3) (D) [0, 4/3)

Ans. D

Sol. $cx^2 + 12 + 6cx > 0$; $c = 0$ is obviously

5. If the vector $6\hat{i} - 3\hat{j} - 6\hat{k}$ is decomposed into vectors parallel and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ then the vectors are :

(A) $-\left(\hat{i} + \hat{j} + \hat{k}\right)$ and $7\hat{i} - 2\hat{j} - 5\hat{k}$ (B) $-2\left(\hat{i} + \hat{j} + \hat{k}\right)$ and $8\hat{i} - \hat{j} - 4\hat{k}$

(C) $+2\left(\hat{i} + \hat{j} + \hat{k}\right)$ and $4\hat{i} - 5\hat{j} - 8\hat{k}$ (D) None

यदि सदिश $6\hat{i} - 3\hat{j} - 6\hat{k}$ को सदिश $\hat{i} + \hat{j} + \hat{k}$ के समानान्तर एवं लम्बवत् घटको में तोड़ दिया गया है तो वो घटक होंगे :

(A) $-\left(\hat{i} + \hat{j} + \hat{k}\right)$ तथा $7\hat{i} - 2\hat{j} - 5\hat{k}$ (B) $-2\left(\hat{i} + \hat{j} + \hat{k}\right)$ तथा $8\hat{i} - \hat{j} - 4\hat{k}$

(C) $+2\left(\hat{i} + \hat{j} + \hat{k}\right)$ तथा $4\hat{i} - 5\hat{j} - 8\hat{k}$ (D) कोई नहीं

Ans. A

Sol. A vector $\vec{a}$ which is decomposed into parallel and perpendicular to the vector $\vec{b}$ is $\left(\frac{\vec{a}\cdot\vec{b}}{\vec{b}^2}\right)\vec{b}$ and $\vec{a}\left(\frac{\vec{a}\cdot\vec{b}}{\vec{b}^2}\right)\vec{b}$ or $\frac{\vec{b}\times\left(\vec{a}\times\vec{b}\right)}{\vec{b}^2}$

6. Let $\vec{r} = \vec{a} + \lambda\vec{\ell}$ and $\vec{r} = \vec{b} + \mu\vec{m}$ be two lines in space where $\vec{a} = 5\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = -\hat{i} + 7\hat{j} + 8\hat{k}$, $\vec{\ell} = -4\hat{i} + \hat{j} - \hat{k}$ and $\vec{m} = 2\hat{i} - 5\hat{j} - 7\hat{k}$ then the p.v. of a point which lies on both of these lines, is :

(A) $\hat{i} + 2\hat{j} + \hat{k}$ (B) $2\hat{i} + \hat{j} + \hat{k}$

(C) $\hat{i} + \hat{j} + 2\hat{k}$ (D) non existent as the lines are skew

माना $\vec{r} = \vec{a} + \lambda\vec{\ell}$ तथा $\vec{r} = \vec{b} + \mu\vec{m}$ दो रेखाऐं समष्टि में है जहाँ $\vec{a} = 5\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = -\hat{i} + 7\hat{j} + 8\hat{k}$, $\vec{\ell} = -4\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{m} = 2\hat{i} - 5\hat{j} - 7\hat{k}$ तो कौनसा बिन्दु दोनों रेखाओं पर स्थित होगा :

(A) $\hat{i} + 2\hat{j} + \hat{k}$ (B) $2\hat{i} + \hat{j} + \hat{k}$

(C) $\hat{i} + \hat{j} + 2\hat{k}$ (D) विद्यमान नहीं है क्योंकि रेखाऐं विसमतलीय है

Ans. A

Sol. $\lambda = \mu = 1$ (point of intersection of two lines)

7. Let OAB be a regular triangle with side length unity (O being the origin). Also M, N are the points of trisection of AB, M being closer to A and N closer to B. Position vectors of A, B, M and N are $\vec{a}$, $\vec{b}$, $\vec{m}$ and $\vec{n}$ respectively. Which of the following hold(s) good?

(A) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{2}{3}$ and $y = \frac{1}{3}$ (B) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{5}{6}$ and $y = \frac{1}{6}$

(C) $\vec{m} \cdot \vec{n}$ equals $\frac{13}{18}$ (D) $\vec{m} \cdot \vec{n}$ equals $\frac{15}{18}$

माना OAB एक समबाहु त्रिभुज है जिसकी भुजा ईकाई है (O मूल बिन्दु है). साथ ही M, N, भुजा AB के समत्रिभाजक बिन्दु है, M बिन्दु A के समीप है तथा N, बिन्दु B के समीप है। A, B, M तथा N के स्थिति सदिश क्रमशः $\vec{a}$, $\vec{b}$, $\vec{m}$ तथा $\vec{n}$ है। निम्न में से कौनसे सही है?

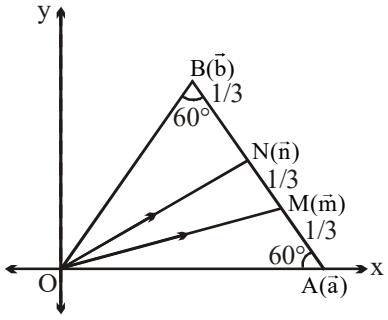
(A) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{2}{3}$ तथा $y = \frac{1}{3}$ (B) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{5}{6}$ तथा $y = \frac{1}{6}$

(C) $\vec{m} \cdot \vec{n} = \frac{13}{18}$ (D) $\vec{m} \cdot \vec{n} = \frac{15}{18}$

Ans. AC

Sol. $\vec{m} = \frac{1}{3}\vec{b} + \frac{2}{3}\vec{a}$ (Using Section formula)

$\Rightarrow \quad x = \frac{2}{3}$ and $y = \frac{1}{3} \quad \Rightarrow \quad$ (A)



$\vec{m} = \frac{2\vec{a} + \vec{b}}{3}$

again $\vec{n} = \frac{1}{3}\vec{a} + \frac{2}{3}\vec{b} = \frac{\vec{a} + 2\vec{b}}{3}$

$\vec{m} \cdot \vec{n} = \left(\frac{2\vec{a} + \vec{b}}{3}\right) \cdot \left(\frac{\vec{a} + 2\vec{b}}{3}\right)$ where $\left(\vec{a} \cdot \vec{b} = \frac{1}{2}\right)$

$= \frac{1}{9}\left[2 + 4\left(\frac{1}{2}\right) + \frac{1}{2} + 2\right] = \frac{13}{18} \Rightarrow$ (C)

8. In ΔABC, a point P is chosen on side $\overrightarrow{AB}$ so that AP : PB = 1 : 4 and a point Q is chosen on the side $\overrightarrow{BC}$ so that CQ : QB = 1 : 3. Segment $\overrightarrow{CP}$ and $\overrightarrow{AQ}$ intersect at M. If the ratio $\frac{MC}{PC}$ is expressed as a rational numberm in the

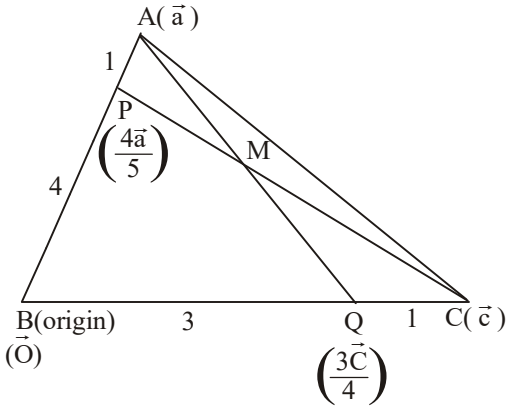
lowest term as $\frac{a}{b}$, find (a + b).

माना ΔABC में, बिन्दु P भुजा $\overrightarrow{AB}$ पर इस प्रकार है कि AP : PB = 1 : 4 है तथा बिन्दु Q भुजा $\overrightarrow{BC}$ पर इस प्रकार है कि CQ : QB = 1 : 3 है। रेखाखण्ड $\overrightarrow{CP}$ तथा $\overrightarrow{AQ}$ बिन्दु M पर प्रतिच्छेद करते है। यदि अनुपात $\frac{MC}{PC}$ को एक परिमेय $\frac{a}{b}$ के न्यूनतम समकक्ष रूप में लिखा गया है तो (a + b) का मान है।

Ans. 13

Sol. Vector equation of AQ is $\vec{r}_1 = \vec{a} + \lambda\left(\vec{a} - \frac{3\vec{c}}{4}\right)$

and vector equation of CP is $\vec{r}_2 = \vec{c} + \mu\left(\vec{c} - \frac{4\vec{a}}{5}\right)$



hence $\vec{r}_1 = \vec{r}_2$ gives, $\left(1 + \lambda + \frac{4\mu}{5}\right)\vec{a} = \left(1 + \mu + \frac{3\lambda}{4}\right)\vec{c}$

$\therefore \quad 1 + \lambda + \frac{4\mu}{5} = 0 \quad$ and $\quad 1 + \mu + \frac{3\lambda}{4} = 0$

solving, $\lambda = -\frac{1}{2} \quad$ and $\quad \mu = -\frac{5}{8}$

$\therefore \quad$ p.v. of $M = \vec{a} - \frac{1}{2}\vec{a} + \frac{3\vec{c}}{8} = \frac{4\vec{a} + 3\vec{c}}{8}$

$$\overrightarrow{MC} = \vec{c} - \frac{4\vec{a} + 3\vec{c}}{8} = \frac{5\vec{c} - 4\vec{a}}{8}$$

and $\quad \overrightarrow{PC} = \vec{c} - \frac{4\vec{a}}{5} = \frac{5\vec{c} - 4\vec{a}}{5}$

$\therefore \quad \frac{MC}{PC} = \frac{5}{8} \quad \Rightarrow \quad a = 5$ and $b = 8; \quad \therefore \; a + b = 13$ Ans.

1. If $\vec{a}+\vec{b}+\vec{c}=0$, $|\vec{a}|=3$, $|\vec{b}|=5$, $|\vec{c}|=7$, then the angle between $\vec{a}$ and $\vec{b}$ is :

यदि $\vec{a}+\vec{b}+\vec{c}=0$, $|\vec{a}|=3$, $|\vec{b}|=5$, $|\vec{c}|=7$, तो $\vec{a}$ तथा $\vec{b}$ के मध्य कोण है :

(A) $\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $\frac{5\pi}{3}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $\vec{a}+\vec{b}=-\vec{c}$ Square $|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\,.\,\vec{b}=|\vec{c}|^2$

$\therefore \quad \cos\theta=\frac{|\vec{c}|^2-|\vec{a}|^2-|\vec{b}|^2}{2|\vec{a}||\vec{b}|}$

2. A line passes through the point $A\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ and is parallel to the vector $\vec{V}=\left(\hat{i}+\hat{j}+\hat{k}\right)$. The shortest distance from the origin, of the line is :

एक रेखा बिन्दु $A\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ से गुजरती है तथा सदिश $\vec{V}=\left(\hat{i}+\hat{j}+\hat{k}\right)$ के समान्तर है। मूल बिन्दु से रेखा की न्यूनतम दूरी है :

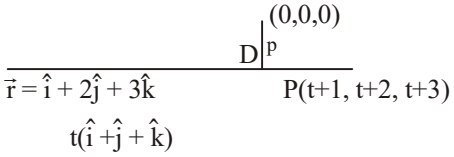
(A) $\sqrt{2}$ (B) $\sqrt{4}$ (C) $\sqrt{5}$ (D) $\sqrt{6}$

Ans. A

Sol. $D^2=f(t)=(t+1)^2+(t+2)^2+(t+3)^2$

$f'(t)=2[(1+t)+2+t+3+t]$

$f'(t)=0 \quad \Rightarrow \quad t=-2$



hence $D^2(\min)=1+0+1=2$

$D|_{min}=\sqrt{2}$ Ans.

3. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be vectors of length 3, 4, 5 respectively. Let $\vec{a}$ be perpendicular to $\left(\vec{b}+\vec{c}\right)$, $\vec{b}$ to $\left(\vec{c}+\vec{a}\right)$ and $\vec{c}$ to $\left(\vec{a}+\vec{b}\right)$. Then $\left|\vec{a}+\vec{b}+\vec{c}\right|$ is :

माना $\vec{a}$, $\vec{b}$, $\vec{c}$ सदिश है जिनकी लम्बाई क्रमशः 3, 4, 5 है। माना $\vec{a}$ सदिश $\left(\vec{b}+\vec{c}\right)$ के लम्बवत् है, $\vec{b}$ सदिश $\left(\vec{c}+\vec{a}\right)$ के तथा $\vec{c}$ सदिश $\left(\vec{a}+\vec{b}\right)$ के लम्बवत् है तो $\left|\vec{a}+\vec{b}+\vec{c}\right|$ का मान है :

(A) $2\sqrt{5}$ (B) $2\sqrt{2}$ (C) $10\sqrt{5}$ (D) $5\sqrt{2}$

Ans. D

Sol. $\left.\begin{array}{l}\vec{a}\,.\left(\vec{b}+\vec{c}\right)=0\\ \vec{b}\,.\left(\vec{c}+\vec{a}\right)=0\\ \vec{c}\,.\left(\vec{a}+\vec{b}\right)=0\end{array}\right\} \Rightarrow \vec{a}\,.\,\vec{b}+\vec{b}\,.\,\vec{c}+\vec{c}\,.\,\vec{a}\;=0$

$\therefore\ \vec{a}\,.\,\vec{b}=\vec{b}\,.\,\vec{c}=\vec{c}\,.\,\vec{a}\;=0$ now square $\left|\vec{a}+\vec{b}+\vec{c}\right|$ to get the result

4. Given a parallelogram ABCD. If $\left|\vec{AB}\right|=a$, $\left|\vec{AD}\right|=b$ and $\left|\vec{AC}\right|=c$, then $\vec{DB}\,.\,\vec{AB}$ has the value :

(A) $\dfrac{3a^2+b^2-c^2}{2}$ (B) $\dfrac{a^2+3b^2-c^2}{2}$ (C) $\dfrac{a^2-b^2+3c^2}{2}$ (D) None

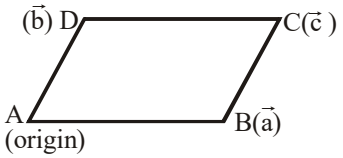
दिये गये समान्तर चतुर्भुज ABCD है, यदि $\left|\vec{AB}\right|=a$, $\left|\vec{AD}\right|=b$ तथा $\left|\vec{AC}\right|=c$ है, तो $\vec{DB}\,.\,\vec{AB}$ का मान है :

(A) $\dfrac{3a^2+b^2-c^2}{2}$ (B) $\dfrac{a^2+3b^2-c^2}{2}$ (C) $\dfrac{a^2-b^2+3c^2}{2}$ (D) कोई नहीं

Ans. A

Sol. To find $\left(\vec{a}-\vec{b}\right).\vec{a}$ i.e. $\left|\vec{a}\right|^2-\vec{a}\,.\,\vec{b}$ —(1)

now $\vec{a}+\vec{b}=\vec{c}\ \Rightarrow \left|\vec{a}\right|^2+\left|\vec{b}\right|^2+2\,\vec{a}\,.\,\vec{b}=\left|\vec{c}\right|^2$ —(2)



substitute the value of $\vec{a}\,.\,\vec{b}$ from (2) in (1)

5. The set of values of x for which the angle between the vectors $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ acute and the angle between the vector $\vec{b}$ and the axis of ordinates is obtuse, is :

x के मानों का समुच्चय जिसके लिये सदिशों $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ तथा $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ के मध्य न्यूनकोण तथा $\vec{b}$ एक कोटि अक्षः के मध्य अधिक कोण है :

(A) $1 < x < 2$ (B) $x > 2$ (C) $x < 1$ (D) $x < 0$

Ans. D

Sol. $\vec{a} \cdot \vec{b} > 0 \Rightarrow 2x^2 - 3x + 1 > 0$ and $x < 0$

6. If a vector $\vec{a}$ of magnitude 50 is collinear with vector $\vec{b} = 6\hat{i} - 8\hat{j} - \frac{15}{2}\hat{k}$ and makes an acute angle with positive z-axis then :

(A) $\vec{a} = 4\vec{b}$ (B) $\vec{a} = -4\vec{b}$ (C) $\vec{b} = 4\vec{a}$ (D) None

यदि एक सदिश $\vec{a}$ का परिमाण 50 है जो सदिश $\vec{b} = 6\hat{i} - 8\hat{j} - \frac{15}{2}\hat{k}$ के साथ समरेखीय है तथा धनात्मक z-अक्ष के साथ न्यूनत कोण बनाता है, तो :

(A) $\vec{a} = 4\vec{b}$ (B) $\vec{a} = -4\vec{b}$ (C) $\vec{b} = 4\vec{a}$ (D) कोई नहीं

Ans. B

Sol. Let $\vec{a} = \lambda\left(6\hat{i} - 8\hat{j} - \frac{15}{2}\hat{k}\right)$ $|\vec{a}| = 50 \Rightarrow \lambda = \pm 4$ but $\vec{a} \cdot \hat{k} > 0 \Rightarrow \lambda = -4$

7. A, B, C and D are four points in a plane with pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ respectively such that $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$. Then for the triangle ABC, D is its :

(A) incentre (B) circumcentre (C) orthocentre (D) centroid

A, B, C तथा D चार बिन्दु एक तल में स्थित है जिनके स्थिति सदिश pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ क्रमशः इस प्रकार है कि $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$ तो D त्रिभुज ABC का :

(A) अन्तः केन्द्र है (B) परिकेन्द्र है (C) लम्बकेन्द्र है (D) केन्द्रक है

Ans. C

Sol. $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = 0$

$\overrightarrow{DA}\cdot\overrightarrow{CB}=0$

$AD \perp BC$

& $\left(\vec{b}-\vec{d}\right)\cdot\left(\vec{c}-\vec{a}\right)=0 \Rightarrow \overrightarrow{DB}\cdot\overrightarrow{AC}=0$

$\Rightarrow \overrightarrow{BD} \perp AC = 0$

8. $\vec{a}$ and $\vec{b}$ are unit vectors inclined to each other at an angle α, $\alpha \in (0, \pi)$ and $\left|\vec{a}+\vec{b}\right| < 1$. Then $\alpha \in$:

$\vec{a}$ तथा $\vec{b}$ इकाई सदिश है जो परस्पर α कोण बनाते है, $\alpha \in (0, \pi)$ तथा $\left|\vec{a}+\vec{b}\right| < 1$ है तो $\alpha \in$:

(A) $\left(\frac{\pi}{3}, \frac{2\pi}{3}\right)$ (B) $\left(\frac{2\pi}{3}, \pi\right)$ (C) $\left(0, \frac{\pi}{3}\right)$ (D) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

Ans. B

Sol. $\left|\vec{a}+\vec{b}\right|^2 < 1 \Rightarrow 2 + 2\cos\alpha < 1$

$\Rightarrow \cos\frac{\alpha}{2} < \frac{-1}{2} \Rightarrow \alpha \in \left(\frac{2\pi}{3}, \pi\right)$ Ans.

9. Image of the point P with position vector $7\hat{i} - \hat{j} + 2\hat{k}$ in the line whose vector equation is, $\vec{r} = 9\hat{i}+5\hat{j}+5\hat{k}+\lambda\left(\hat{i}+3\hat{j}+5\hat{k}\right)$ has the position vector :

(A) $(-9, 5, 2)$ (B) $(9, 5, -2)$ (C) $(9, -5, -2)$ (D) None

बिन्दु P जिसका स्थिति सदिश $7\hat{i} - \hat{j} + 2\hat{k}$ है का रेखा, जिसका सदिश समीकरण $\vec{r} = 9\hat{i}+5\hat{j}+5\hat{k}+\lambda\left(\hat{i}+3\hat{j}+5\hat{k}\right)$ है, में प्रतिबिम्ब है :

(A) $(-9, 5, 2)$ (B) $(9, 5, -2)$ (C) $(9, -5, -2)$ (D) कोई नहीं

Ans. B

Sol. $\frac{x-9}{1} = \frac{y-5}{3} = \frac{z-5}{5}$

P is on line $\Rightarrow \frac{\alpha+7-18}{2} = \frac{-1+\beta-10}{6} = \frac{\gamma+2-10}{10}$

$\Rightarrow \alpha - 11 = \frac{\beta-11}{3} = \frac{\gamma-8}{5}$

$(7-\alpha)+3(-1-\beta)+5(2-\gamma)=0$

$\alpha+3\beta+5\gamma=14$ $\qquad\qquad$ $\alpha=\lambda+11,\ \beta=3\lambda+11,\ \gamma=5\lambda+8$

$\lambda+11+9\lambda+33+25\lambda+40=14$

$35\lambda+70=0\Rightarrow\lambda=-2$

$(\alpha,\beta,\gamma)=(9,5,-2)$

10. Let $\hat{a}$, $\hat{b}$, $\hat{c}$ are three unit vectors such that $\hat{a}+\hat{b}+\hat{c}$ is also a unit vector. If pairwise angles between $\hat{a}$, $\hat{b}$, $\hat{c}$ are θ_1, θ_2 and θ_3 rexpectively then $\cos\theta_1+\cos\theta_2+\cos\theta_3$ equals :

माना $\hat{a}$, $\hat{b}$, $\hat{c}$ तीन इकाई सदिश इस प्रकार है कि $\hat{a}+\hat{b}+\hat{c}$ भी एक इकाई सदिश है। यदि $\hat{a}$, $\hat{b}$, $\hat{c}$ के मध्य युग्मों में कोण θ_1, θ_2 तथा θ_3 क्रमशः है तो $\cos\theta_1+\cos\theta_2+\cos\theta_3$ का मान हैं :

(A) 3 (B) -3 (C) 1 (D) -1

Ans. D

Sol. consider $\left(\hat{a}+\hat{b}+\hat{c}\right)^2=3+2(\cos\theta_1+\cos\theta_2+\cos\theta_3)\Rightarrow 1=3+2\sum\cos\theta_i\ \Rightarrow\ \sum\cos\theta_i=-1$

11. L_1 and L_2 are two lines whose vector equations are

$L_1:\ \vec{r}=\lambda\left[\left(\cos\theta+\sqrt{3}\right)\hat{i}+\left(\sqrt{2}\sin\theta\right)\hat{j}+\left(\cos\theta-\sqrt{3}\right)\hat{k}\right]$

$L_2:\ \vec{r}=\mu\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$,

where λ and μ are scalars and α is the acute angle between L_1 and L_2. If the angle 'α' is independent of θ then the value of 'α' is :

L_1 तथा L_2 दो रेखाओं का सदिश समीकरण निम्न प्रकार से है

$L_1:\ \vec{r}=\lambda\left[\left(\cos\theta+\sqrt{3}\right)\hat{i}+\left(\sqrt{2}\sin\theta\right)\hat{j}+\left(\cos\theta-\sqrt{3}\right)\hat{k}\right]$

$L_2:\ \vec{r}=\mu\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$,

जहाँ λ तथा μ सदिश है तथा L_1 तथा L_2 के मध्य न्यूनकोण α है। यदि कोण 'α', θ से स्वतंत्र है तो 'α' का मान है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. A

Sol. Both the lines pass through origin

Line L_1 is parallel to the vector

$$\vec{V}_1=\left(\cos\theta+\sqrt{3}\right)\hat{i}+\left(\sqrt{2}\sin\theta\right)\hat{j}+\left(\cos\theta-\sqrt{3}\right)\hat{k}$$

and L_2 is parallel to the vector

$$\vec{V}_2=a\hat{i}+b\hat{j}+c\hat{k}$$

$$\therefore \qquad \cos\alpha = \frac{\vec{V}_1 \cdot \vec{V}_2}{|\vec{V}_1||\vec{V}_2|} = \frac{a(\cos\theta+\sqrt{3})+(b\sqrt{2})\sin\theta+c(\cos\theta-\sqrt{3})}{\sqrt{a^2+b^2+c^2}\sqrt{(\cos\theta+\sqrt{3})^2+2\sin^2\theta+(\cos\theta-\sqrt{3})^2}}$$

$$= \frac{(a+c)\cos\theta+b\sqrt{2}\sin\theta+(a-c)\sqrt{3}}{\sqrt{a^2+b^2+c^2}\ \sqrt{2+6}}$$

in order that $\cos\alpha$ is independent of θ

$a + c = 0$ and $b = 0$

$$\therefore \qquad \cos\alpha = \frac{2a\sqrt{3}}{a\sqrt{2}\cdot 2\sqrt{2}} = \frac{\sqrt{3}}{2} \qquad \Rightarrow \qquad \alpha = \frac{\pi}{6} \text{ Ans.}$$

1. Cosine of an angle between the vectors $\left(\vec{a}+\vec{b}\right)$ and $\left(\vec{a}-\vec{b}\right)$ if $|\vec{a}|=2$, $|\vec{b}|=1$ and $\vec{a}\wedge\vec{b}=60°$ is :

(A) $\sqrt{\frac{3}{7}}$ (B) $\frac{9}{\sqrt{21}}$ (C) $\frac{3}{\sqrt{7}}$ (D) None

सदिशो $\left(\vec{a}+\vec{b}\right)$ तथा $\left(\vec{a}-\vec{b}\right)$, यदि $|\vec{a}|=2$, $|\vec{b}|=1$, के मध्य कोण की कोज्या बताइए यदि $\vec{a}\wedge\vec{b}=60°$ है :

(A) $\sqrt{\frac{3}{7}}$ (B) $\frac{9}{\sqrt{21}}$ (C) $\frac{3}{\sqrt{7}}$ (D) कोई नहीं

Ans. A

Sol. $\cos\theta=\frac{(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})}{|\vec{a}+\vec{b}|\left|\vec{a}-\vec{b}\right|}=\frac{\vec{a}^2-\vec{b}^2}{|\vec{a}+\vec{b}|\left|\vec{a}-\vec{b}\right|}=\frac{3}{|\vec{V}_1|\left|\vec{V}_2\right|}$

2. An arc AC of a circle subtends a right angle at the centre O. The point B divides the arc in the ratio 1 : 2. If $\vec{OA}=\vec{a}$ and $\vec{OB}=\vec{b}$, then the vector $\overrightarrow{OC}$ in terms of $\vec{a}$ and $\vec{b}$, is :

AC एक वृत पर चाप है जो केन्द्र O पर समकोण अंकित करता है। बिन्दु B उस चाप को 1 : 2 में विभाजित करता है। यदि $\vec{OA}=\vec{a}$ तथा $\vec{OB}=\vec{b}$ तो सदिश $\overrightarrow{OC}$, $\vec{a}$ तथा $\vec{b}$ के पदों में है :

(A) $\sqrt{3}\,\vec{a}-2\vec{b}$ (B) $-\sqrt{3}\,\vec{a}+2\vec{b}$ (C) $2\vec{a}-\sqrt{3}\,\vec{b}$ (D) $-2\vec{a}+\sqrt{3}\,\vec{b}$

Ans. B

Sol. $\vec{c}=\lambda\vec{a}+\mu\vec{b}$

dot with $\vec{a}$

$0=\lambda|\vec{a}|+\mu\vec{a}\cdot\vec{b}$

$2\lambda+\sqrt{3}\mu=0\Rightarrow\lambda=-\sqrt{3}$

dot with $\vec{c}\Rightarrow$ $|\vec{c}|^2=\mu\vec{b}\cdot\vec{c}\Rightarrow\mu=2$

$\vec{c}=-\sqrt{3}\vec{a}+2\vec{b}$

3. Given three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ each two of which are non collinear. Further if $\left(\vec{a}+\vec{b}\right)$ is collinear with $\vec{c}$, $\left(\vec{b}+\vec{c}\right)$ is collinear with $\vec{a}$ and $|\vec{a}|=|\vec{b}|=|\vec{c}|=\sqrt{2}$. Then the value of $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$:

(A) is 3 (B) is -3 (C) is 0 (D) cannot be evaluated

दिया गया $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ सदिश है जिनमें से कोई दो सदिश समरेखीय नहीं है। साथ ही $\left(\vec{a}+\vec{b}\right)$ सदिश $\vec{c}$ के साथ, $\left(\vec{b}+\vec{c}\right)$ सदिश $\vec{a}$ के साथ समरेखीय $|\vec{a}|=|\vec{b}|=|\vec{c}|=\sqrt{2}$ है, तो $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ है :

(A) is 3 (B) is -3 (C) is 0 (D) ज्ञात नहीं किया जा सकता

Ans. B

Sol. $\vec{a}+\vec{b}=\lambda\vec{c}$ ....(1)

and $\vec{b}+\vec{c}=\mu\vec{a}$ ....(2)

$\therefore$ $(\lambda\vec{c}-\vec{a})+\vec{c}=\mu\vec{a}$ [putting $\vec{b}=\lambda\vec{c}-\vec{a}$]

$(\lambda+1)\ \vec{c}=(\mu+1)\ \vec{a}$ $\Rightarrow$ $\lambda=\mu=-1$

$\therefore$ $\vec{a}+\vec{b}+\vec{c}=0$ now proceed

4. The vector equations of two lines L_1 and L_2 are respectivly

$\vec{r}=17\hat{i}-9\hat{j}+9\hat{k}+\lambda(3\hat{i}+\hat{j}+5\hat{k})$ and $\vec{r}=15\hat{i}-8\hat{j}-\hat{k}+\mu(4\hat{i}+3\hat{j})$

I L_1 and L_2 are skew lines

II $(11,-11,-1)$ is the point of intersection of L_1 and L_2

III $(-11, 11, 1)$ is the point of intersection of L_1 and L_2

IV $\cos^{-1}\left(\frac{3}{\sqrt{35}}\right)$ is the acute angle between L_1 and L_2

then , which of the following is true?

(A) II and IV (B) I and IV (C) IV only (D) III and IV

रेखाओं L_1 तथा L_2 के क्रमशः समीकरण है

$\vec{r}=17\hat{i}-9\hat{j}+9\hat{k}+\lambda(3\hat{i}+\hat{j}+5\hat{k})$ तथा $\vec{r}=15\hat{i}-8\hat{j}-\hat{k}+\mu(4\hat{i}+3\hat{j})$

I L_1 तथा L_2 विसमतलीय रेखायें है

II L_1 तथा L_2 का प्रतिच्छेदन बिन्दु $(11, -11, -1)$ है

III L_1 तथा L_2 का प्रतिच्छेदन बिन्दु $(-11, 11, 1)$ है

IV L_1 तथा L_2 के मध्य न्यून कोण $\cos^{-1}\left(\frac{3}{\sqrt{35}}\right)$ है

तो उपर्युक्त में से कौन सही है?

(A) II तथा IV (B) I तथा IV (C) केवल IV (D) III तथा IV

Ans. A

Sol. $\vec{r} = (17,-9,9) + \lambda(3,1,5)$

$\vec{r} = (15,-8,-1) + \mu(4,3,0)$

Now Point of intersection $\Rightarrow 17 + 3\lambda = 15 + 4\mu$

$9 + 5\lambda = -1 \Rightarrow \lambda = -2, \mu = -1$

Verify $-9 + \lambda = -8 + 3\mu \Rightarrow -11 = -11$

Point of intersection $(11, -11, -1)$

(2) $\cos\theta = \frac{3\times4+1\times3}{\sqrt{9+1+25}\sqrt{16+9}} = \frac{3}{\sqrt{35}}$

5. For two particular vectors $\vec{A}$ and $\vec{B}$ it is known that $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$. What must be true about the two vectors?

(A) At least one of the two vectors must be the zero vector

(B) $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$ is true for any two vectors

(C) One of the two vectors is a scalar multiple of the other vector

(D) The two vectors must be perpendicular to each other

दो सदिश $\vec{A}$ तथा $\vec{B}$ के लिये ज्ञात है कि $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$ है। निम्न में से क्या हमेशा सत्य है?

(A) कम से कम एक सदिश शून्य है

(B) $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$ किन्ही दो सदिशों के लिये सत्य है

(C) दोनों में से एक सदिश अन्य सदिश का अदिश गुणंक है

(D) दोनों सदिश लम्बवत् है

Ans. C

Sol. $\vec{A}\times\vec{B} = -\vec{A}\times\vec{B}$

$\vec{A}\times\vec{B} = 0$

either $\vec{A} = 0$ or $\vec{B} = 0$

or $\vec{A}$ and $\vec{B}$ are collinear

6. For some non zero vector $\vec{V}$, if the sum of $\vec{V}$ and the vector obtained from $\vec{V}$ by rotating it by an angle 2α equals to the vector obtained from $\vec{V}$ by rotating it by α then the values of α, is : (where n is an integer)

किसी अशून्य सदिश $\vec{V}$ के लिये, यदि $\vec{V}$ तथा इसे 2α से घुर्णन करने के पश्चात् मिले सदिश के योग, एवं $\vec{V}$ को α से घुर्णन करने के पश्चात् मिले सदिश के समान है तो α का मान है : (जहाँ n पूर्णांक है)

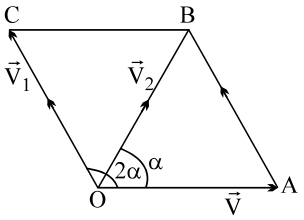
(A) $2n\pi \pm \frac{\pi}{3}$ (B) $n\pi \pm \frac{\pi}{3}$ (C) $2n\pi \pm \frac{2\pi}{3}$ (D) $n\pi \pm \frac{2\pi}{3}$

Ans. A

Sol. Given $\vec{V} + \vec{V}_1 = \vec{V}_2$; Also $\vec{V} \wedge \vec{V}_1 = 2\alpha$

$(\vec{V}_1)^2 = (\vec{V}_2 - \vec{V})^2$ and $\vec{V} \wedge \vec{V}_2 = \alpha$

also, $|\vec{V}| = |\vec{V}_1| = |\vec{V}_2| = \lambda$ say



hence, $\lambda^2 = 2\lambda^2 - 2\lambda^2 \cos\alpha \Rightarrow \cos\alpha = 1/2 \Rightarrow \alpha = 2n\pi \pm (\pi/3)$ Ans.

7. Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}| = 1$, $|\vec{v}| = 2$, $|\vec{w}| = 3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and vectors $\vec{v}$, $\vec{w}$ are perpendicular to each other then $|\vec{u} - \vec{v} + \vec{w}|$ equals :

माना $\vec{u}, \vec{v}, \vec{w}$ इस प्रकार है कि $|\vec{u}| = 1$, $|\vec{v}| = 2$, $|\vec{w}| = 3$ यदि $\vec{v}$ का $\vec{u}$ पर प्रक्षेप $\vec{w}$ का $\vec{u}$ पर प्रक्षेप के बराबर है तथा $\vec{v}$, $\vec{w}$ लम्बवत् है तो $|\vec{u} - \vec{v} + \vec{w}|$ का मान है :

(A) 2 (B) $\sqrt{7}$ (C) $\sqrt{14}$ (D) 14

Ans. C

Sol. $\vec{v} \cdot \hat{u} = \vec{w} \cdot \hat{u}$

$\vec{v} \perp \vec{w} \Rightarrow \vec{v} \cdot \vec{w} = 0$

now, $|\vec{u} - \vec{v} + \vec{w}|^2 = \vec{u}^2 + \vec{v}^2 + \vec{w}^2 - 2\vec{u} \cdot \vec{v} - 2\vec{w} \cdot \vec{v} + 2\vec{u} \cdot \vec{w} = 1 + 4 + 9$

so $|\vec{u} - \vec{v} + \vec{w}| = \sqrt{14}$ Ans.

8. If $\vec{a}$ and $\vec{b}$ are non zero, non collinear vectors, and the linear combination $(2x - y)\vec{a} + 4\vec{b} = 5\vec{a} + (x - 2y)\vec{b}$ holds for real x and y then x + y has the value equal to :

यदि $\vec{a}$ तथा $\vec{b}$ अशून्य, विषमरेखीय सदिश है तथा रेखीय संचय $(2x - y)\vec{a} + 4\vec{b} = 5\vec{a} + (x - 2y)\vec{b}$ किसी x तथा y के लिये है, तो x + y का मान है :

(A) – 3 (B) 1 (C) 17 (D) 3

Ans. B

Sol. $(2x-y-5)\vec{a}=(x-2y-4)\vec{b}$

$\therefore \quad 2x-y=5 \quad ....(1); \qquad x-2y=4 \quad ....(2)$

from (1) and (2)

$2(2y+4)-y=5 \quad \Rightarrow \quad 3y=-3 \quad \Rightarrow \quad y=-1$

and $x=2;$ hence $x+y=1$ Ans.

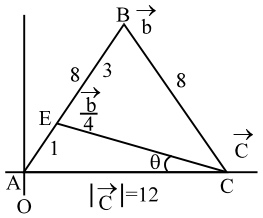
9. In the isosceles triangle ABC $\left|\vec{AB}\right|=\left|\vec{BC}\right|=8$, a point E divides AB internally in the ratio 1 : 3, then the cosine of the angle between $\vec{CE}$ and $\vec{CA}$ is : (where $\left|\vec{CA}\right|=12$)

एक समद्विबाहु त्रिभुज ABC $\left|\vec{AB}\right|=\left|\vec{BC}\right|=8$ है, बिन्दु E, AB को 1 : 3 में अन्तः विभाजन करता है तो $\vec{CE}$ तथा $\vec{CA}$ के कोण की कोज्या है : (जहाँ $\left|\vec{CA}\right|=12$)

(A) $-\frac{3\sqrt{7}}{8}$ (B) $\frac{3\sqrt{8}}{17}$ (C) $\frac{3\sqrt{7}}{8}$ (D) $\frac{-3\sqrt{8}}{17}$

Ans. C

Sol. Given $\left|\vec{b}\right|=\left|\vec{b}-\vec{c}\right|=8$ and $\left|\vec{c}\right|=12$; $\cos\theta=\frac{\vec{c}.\left(\vec{c}-\frac{\vec{b}}{4}\right)}{\left|\vec{c}\right|\left|\vec{c}-\frac{\vec{b}}{4}\right|}=\frac{\vec{c}^2-\frac{\vec{c}.\vec{b}}{4}}{12\left|\vec{c}-\frac{\vec{b}}{4}\right|}$



now proceed.

10. If $\vec{p}=3\vec{a}-5\vec{b}$; $\vec{q}=2\vec{a}+\vec{b}$; $\vec{r}=\vec{a}+4\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ are four vectors such that $\sin\left(\vec{p}\ \wedge\ \vec{q}\right)=1$ and $\sin\left(\vec{r}\ \wedge\ \vec{s}\right)=1$ then $\cos\left(\vec{a}\ \wedge\ \vec{b}\right)$ is :

यदि $\vec{p}=3\vec{a}-5\vec{b}$; $\vec{q}=2\vec{a}+\vec{b}$; $\vec{r}=\vec{a}+4\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ चार सदिश इस प्रकार है कि $\sin\left(\vec{p}\ \wedge\ \vec{q}\right)=1$ तथा $\sin\left(\vec{r}\ \wedge\ \vec{s}\right)=1$ तो $\cos\left(\vec{a}\ \wedge\ \vec{b}\right)$ is :

(A) $-\frac{19}{5\sqrt{43}}$ (B) 0 (C) 1 (D) $\frac{19}{5\sqrt{43}}$

Ans. D

Sol. $\vec{p}\,.\,\vec{q} = 0$ & $\vec{r}\,.\,\vec{s} = 0$. Form two simultaneous equations and get a relation between $|\vec{a}|$ & $|\vec{b}|$ as $25|\vec{a}|^2 = 43|\vec{b}|^2$. Now compute $\cos(\vec{a} \wedge \vec{b}) = \dfrac{\vec{a}\,.\,\vec{b}}{|\vec{a}|\,|\vec{b}|}$

11. Given an equilateral triangle ABC with side length equal to 'a'. Let M and N be two points respectively on the side AB and AC much that $\overrightarrow{AN} = K\overrightarrow{AC}$ and $\overrightarrow{AM} = \dfrac{\overrightarrow{AB}}{3}$. If $\overrightarrow{BN}$ and $\overrightarrow{CM}$ are orthogonal then the value of K is equal to :

दिया गया है कि ABC एक समबाहु त्रिभुज की जिसकी भुजा 'a' है। मान M तथा N दो बिन्दु क्रमशः भुजा AB तथा AC पर है कि $\overrightarrow{AN} = K\overrightarrow{AC}$ तथा $\overrightarrow{AM} = \dfrac{\overrightarrow{AB}}{3}$ है। यदि $\overrightarrow{BN}$ तथा $\overrightarrow{CM}$ लम्बवत् है तो K का मान है :

(A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Ans. A

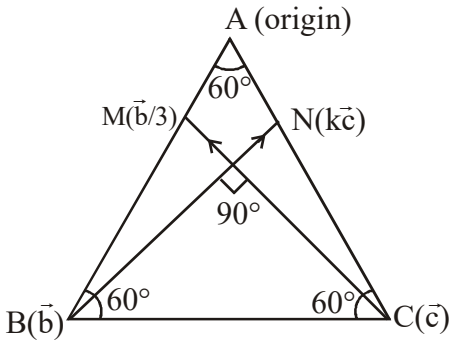
Sol. $\overrightarrow{BN} \cdot \overrightarrow{CM} = 0$

$$\left(k\vec{c} - \vec{b}\right) \cdot \left(\frac{\vec{b}}{3} - \vec{c}\right) = 0$$

$$\frac{k}{3} \cdot \frac{a^2}{2} - ka^2 - \frac{a^2}{3} + \frac{a^2}{2} = 0$$

$$\frac{k}{6} - k + \frac{1}{6} = 0$$

$$\frac{5k}{6} = \frac{1}{6} \qquad \Rightarrow \qquad k = \frac{1}{5} \text{ Ans.}$$



12. Let $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u} \cdot \hat{n} = 0$ and $\vec{v} \cdot \hat{n} = 0$, then $|\vec{w} \cdot \hat{n}|$ is equal to :

माना $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ तथा $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$ है। यदि $\hat{n}$ इस प्रकार है कि $\vec{u} \cdot \hat{n} = 0$ तथा $\vec{v} \cdot \hat{n} = 0$ है, तो $|\vec{w} \cdot \hat{n}|$ बराबर है :

(A) 1 (B) 2 (C) 3 (D) 0

Ans. C

Sol. $\hat{n} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, where $a_1^2 + a_2^2 + a_3^2 = 1$

$\vec{u}\cdot\hat{n} = 0 \Rightarrow a_1 + a_2 = 0$

also $\vec{v}\cdot\hat{n} = 0 \Rightarrow a_1 - a_2 = 0$

hence, $a_1 = a_2 = 0$

$\therefore$ $a_3 = 1$ or -1

$\therefore$ $\hat{n} = \hat{k}$ or $-\hat{k}$

$|\vec{w}\cdot\hat{n}| = 3$

13. Match the following Columns :

Column I	**Column II**
(A) P is point in the plane of the triangle ABC. pv's of A, B and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively with respect to P as the origin. If $(\vec{b}+\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ and $(\vec{c}+\vec{a})\cdot(\vec{c}-\vec{a}) = 0$, then w.r.t. the triangle ABC, P is its	(P) centroid
(B) If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the three non collinear points A, B and C respectively such that the vector $\vec{V} = \overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC}$ is a null vector then w.r.t. the ΔABC, P is its	(Q) orthocentre
(C) If P is a point inside the Δ ABC such that the vector $\vec{R} = (BC)(\overrightarrow{PA}) + (CA)(\overrightarrow{PB}) + (AB)(\overrightarrow{PC})$ is a null vector then w.r.t. the ΔABC, P is its	(R) Incentre
(D) If P is a point in the plane of the triangle ABC such that the scalar product $\overrightarrow{PA}\cdot\overrightarrow{CB}$ and $\overrightarrow{PB}\cdot\overrightarrow{AC}$ vanishes, then w.r.t. the ΔABC, P is its	(S) circumcentre

निम्न Columns को सुमेलित कीजिए :

Column I	**Column II**
(A) त्रिभुज ABC के तल में बिन्दु P स्थित है। A, B तथा C के स्थिति सदिश क्रमशः $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ है जहाँ P मूल बिन्दु है। यदि $(\vec{b}+\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ तथा $(\vec{c}+\vec{a})\cdot(\vec{c}-\vec{a}) = 0$ तो त्रिभुज ABC	(P) केन्द्रक

त्रिभुज ABC के सापेक्ष P है

(B) यदि $\vec{a}, \vec{b}, \vec{c}$ तीन विषमरेखीय बिन्दुओं A, B तथा C के स्थिति सदिश है तो $\vec{V} = \overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC}$ एक शून्य सदिश है तो त्रिभुज ABC के सापेक्ष P है — (Q) लम्बकेन्द्र

(C) यदि P त्रिभुज ABC के अन्दर एक बिन्दु इस प्रकार है कि सदिश $\vec{R} = (BC)(\overrightarrow{PA}) + (CA)(\overrightarrow{PB}) + (AB)(\overrightarrow{PC})$ एक शून्य सदिश तो ΔABC के सापेक्ष P है — (R) अन्तःकेन्द्र

(D) यदि P बिन्दु त्रिभुज ABC के तल में स्थित है इस प्रकार की अदिश गुणनफल $\overrightarrow{PA} \cdot \overrightarrow{CB}$ तथा $\overrightarrow{PB} \cdot \overrightarrow{AC}$ शून्य हो जाता है, तो त्रिभुज ABC के सापेक्ष P है — (S) परिकेन्द्र

Ans. (A) S; (B) P; (C) R; (D) Q

Sol. (A) $\left|\vec{b}\right| = \left|\vec{c}\right| = \left|\vec{a}\right| \Rightarrow \left|\overrightarrow{PA}\right| = \left|\overrightarrow{PB}\right| = \left|\overrightarrow{PC}\right|$ (S) Circumcentre

(B) $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = 0 \Rightarrow \left(\vec{a} + \vec{b} + \vec{c}\right) - 3\vec{P} = 0$ (P) Centroid

(C) $\gamma\overrightarrow{PA} + S\left(\overrightarrow{PB}\right) + t\left(\overrightarrow{PC}\right) = 0$

$\vec{P} = \dfrac{r\vec{a} + s\vec{b} + t\vec{c}}{r + s + t}$ (R) In centre

(D) $PA \perp CB$ & $PB \perp AC$ (Q) Orthocentre

1. If $\vec{p}$ and $\vec{s}$ are not perpendicular to each other and $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ and $\vec{r} \cdot \vec{s} = 0$, then $\vec{r}$ =

(A) $\vec{p} \cdot \vec{s}$ (B) $\vec{q} + \left(\frac{\vec{q} \cdot \vec{p}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$

(C) $\vec{q} - \left(\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$ (D) $\vec{q} + \mu\vec{p}$ for all scalars μ

यदि $\vec{p}$ तथा $\vec{s}$ परस्पर लम्बवत् नहीं है तथा $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ तथा $\vec{r} \cdot \vec{s} = 0$ है, तो $\vec{r}$ =

(A) $\vec{p} \cdot \vec{s}$ (B) $\vec{q} + \left(\frac{\vec{q} \cdot \vec{p}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$

(C) $\vec{q} - \left(\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$ (D) $\vec{q} + \mu\vec{p}$ सभी अदिशों μ के लिए

Ans. C

Sol. $(\vec{r} - \vec{q}) \times \vec{p} = 0 \Rightarrow \vec{r} - \vec{q} = \lambda\vec{p} \quad \vec{r} = \lambda\vec{p} + \vec{q}$ — (1) now $\vec{r} \cdot \vec{s} = 0$

$\Rightarrow (\lambda\vec{p} + \vec{q}) \cdot \vec{s} = 0 \quad \lambda = -\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}}$ — (2) Put λ from (2) in (1) to get the result

2. If $\vec{u}$ and $\vec{v}$ are two vectors such that $|\vec{u}| = 3$; $|\vec{v}| = 2$ and $|\vec{u} \times \vec{v}| = 6$ then the correct statement is :

यदि $\vec{u}$ तथा $\vec{v}$ दो सदिश इस प्रकार है कि $|\vec{u}| = 3$; $|\vec{v}| = 2$ तथा $|\vec{u} \times \vec{v}| = 6$ तो कौनसा कथन सही है :

(A) $\vec{u} \wedge \vec{v} \in (0, 90°)$ (B) $\vec{u} \wedge \vec{v} \in (90°, 180°)$ (C) $\vec{u} \wedge \vec{v} = 90°$ (D) $(\vec{u} \times \vec{v}) \times \vec{u} = 6\vec{v}$

Ans. C

Sol. $|\vec{u} \times \vec{v}|^2 = \vec{u}^2\vec{v}^2 - (\vec{u} \cdot \vec{v})^2$

$36 = (9)(4) - (\vec{u} \cdot \vec{v})^2$

$\Rightarrow \quad \vec{u} \cdot \vec{v} = 0 \quad \Rightarrow \quad \vec{u}$ and $\vec{v}$ are orthogonal

also $(\vec{u} \times \vec{v}) \times \vec{u} = (\vec{u} \cdot \vec{u})\vec{v} - (\vec{v} \cdot \vec{u})\vec{u} = 9\vec{v} \Rightarrow$ (D) is incorrect

3. Given a parallelogram OACB. The lengths of the vectors $\vec{OA}$, $\vec{OB}$ and $\vec{AB}$ are a, b and c respectively. The scalar product of the vectors $\vec{OC}$ and $\vec{OB}$ is :

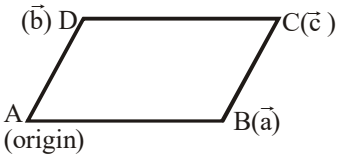
एक समान्तर चतुर्भुज OACB है। सदिशों $\vec{OA}$, $\vec{OB}$ तथा $\vec{AB}$ की लम्बाई क्रमशः a, b तथा c है। $\vec{OC}$ तथा $\vec{OB}$ का अदिश गुणन है :

(A) $\frac{a^2-3b^2+c^2}{2}$ (B) $\frac{3a^2+b^2-c^2}{2}$ (C) $\frac{3a^2-b^2+c^2}{2}$ (D) $\frac{a^2+3b^2-c^2}{2}$

Ans. D

Sol. To find $\left(\vec{a}-\vec{b}\right).\vec{a}$ i.e. $|\vec{a}|^2-\vec{a}.\vec{b}$ —(1)

now $\vec{a}+\vec{b}=\vec{c} \Rightarrow |\vec{a}|^2+|\vec{b}|^2+2\vec{a}.\vec{b}=|\vec{c}|^2$ —(2)



substitute the value of $\vec{a}.\vec{b}$ from (2) in (1)

4. Vectors $\vec{a}$ and $\vec{b}$ make an angle $\theta=\frac{2\pi}{3}$. If $|\vec{a}|=1$, $|\vec{b}|=2$ then $\left\{\left(\vec{a}+3\vec{b}\right)\text{ x }\left(3\vec{a}-\vec{b}\right)\right\}^2=$

सदिश $\vec{a}$ तथा $\vec{b}$ के मध्य कोण $\theta=\frac{2\pi}{3}$ है। यदि $|\vec{a}|=1$, $|\vec{b}|=2$ तो $\left\{\left(\vec{a}+3\vec{b}\right)\text{ x }\left(3\vec{a}-\vec{b}\right)\right\}^2=$

(A) 225 (B) 250 (C) 275 (D) 300

Ans. D

Sol. $100\left|\vec{a}\times\vec{b}\right|^2=300$

5. In a quadrilateral ABCD, $\vec{AC}$ is the bisector of the $\left(\vec{AB}\,\hat{}\,\vec{AD}\right)$ which is $\frac{2\pi}{3}$, $15\left|\vec{AC}\right|=3\left|\vec{AB}\right|=5\left|\vec{AD}\right|$ then $\cos\left(\vec{BA}\,\hat{}\,\vec{CD}\right)$ is :

एक चतुर्भुज ABCD में, $\vec{AC}$, $\left(\vec{AB}\,\hat{}\,\vec{AD}\right)$ का अर्धक है जो $\frac{2\pi}{3}$ है। $15\left|\vec{AC}\right|=3\left|\vec{AB}\right|=5\left|\vec{AD}\right|$ तो $\cos\left(\vec{BA}\,\hat{}\,\vec{CD}\right)$ है :

(A) $-\frac{\sqrt{14}}{7\sqrt{2}}$ (B) $-\frac{\sqrt{21}}{7\sqrt{3}}$ (C) $\frac{2}{\sqrt{7}}$ (D) $\frac{2\sqrt{7}}{14}$

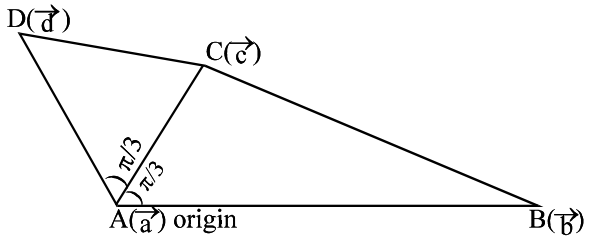
Ans. C

Sol. Given $15|\overrightarrow{AC}| = 3|\overrightarrow{AB}| = 5|\overrightarrow{AD}|$

Let $|\overrightarrow{AC}| = \lambda > 0$

$\therefore |\overrightarrow{AB}| = 5\lambda$

$|\overrightarrow{AD}| = 3\lambda$



Now $\cos\left(\overrightarrow{BA} \wedge \overrightarrow{CD}\right) = \frac{\overrightarrow{BA}.\overrightarrow{CD}}{\left|\overrightarrow{BA}\right|\left|\overrightarrow{CD}\right|} = \frac{\vec{b}.(\vec{d}-\vec{c})}{|\vec{b}||\vec{d}-\vec{c}|}$ .....(1)

Now numerator of $(1) = \vec{b}.\vec{c} - \vec{b}.\vec{d} = |\vec{b}||\vec{c}|\cos\frac{\pi}{3} - |\vec{b}||\vec{d}|\cos\frac{2\pi}{3}$

$$= (5\lambda)(\lambda)\frac{1}{2} + 5\lambda(3\lambda)\frac{1}{2} = \frac{5\lambda^2+15\lambda^2}{2} = 10\lambda^2$$

Denominator of $(1) = |\vec{b}|\,|\vec{d}-\vec{c}|$

Now $|\vec{d}-\vec{c}|^2 = \vec{d}^2 + \vec{c}^2 - 2\vec{c}.\vec{d} = 9\lambda^2 + \lambda^2 - 2(\lambda)(3\lambda)1/2 = 10\lambda^2 - 3\lambda^2 = 7\lambda^2$

$\therefore |\vec{d}-\vec{c}| = \sqrt{7}\,\lambda$

Denominator of $(1) = (5\lambda)(\sqrt{7}\,\lambda) = 5\sqrt{7}\,\lambda^2$

$\therefore \cos\left(\overrightarrow{BA} \wedge \overrightarrow{CD}\right) = \frac{10\lambda^2}{5\sqrt{7}\,\lambda^2} = \frac{2}{\sqrt{7}}$ Ans.

6. If the two adjacent sides of two rectangles are represented by the vectors $\vec{p} = 5\vec{a} - 3\vec{b}$; $\vec{q} = -\vec{a} - 2\vec{b}$ and

$\vec{r} = -4\vec{a} - \vec{b}$; $\vec{s} = -\vec{a} + \vec{b}$ respectively, then the angle between the vectors $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s})$ and $\vec{y} = \frac{1}{5}(\vec{r} + \vec{s})$:

(A) is $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (B) is $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(C) is $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (D) cannot be evaluated

यदि दो आयतो की दो भुजाऐं क्रमशः सदिश $\vec{p} = 5\vec{a} - 3\vec{b}$; $\vec{q} = -\vec{a} - 2\vec{b}$ तथा $\vec{r} = -4\vec{a} - \vec{b}$; $\vec{s} = -\vec{a} + \vec{b}$ द्वारा निरूपित की जाती है तो सदिशो $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s})$ तथा $\vec{y} = \frac{1}{5}(\vec{r} + \vec{s})$ के मध्य कोण है :

(A) is $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (B) is $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(C) is $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (D) ज्ञात नही किया जा सकता

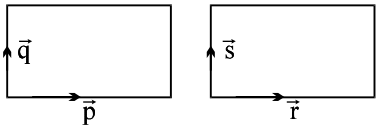
Ans. B

Sol. $\vec{p}.\vec{q} = 0$

$(5\vec{a} - 3\vec{b}).(-\vec{a} - 2\vec{b}) = 0$

$-5\vec{a}^2 - 10\vec{a}\,\vec{b} + 3\vec{a}\,\vec{b} + 6\vec{b}^2 = 0$

$6\vec{b}^2 - 5\vec{a}^2 - 7\vec{a}.\vec{b} = 0$ ....(1)



Also $\vec{r}.\vec{s} = 0$

$(-4\vec{a} - \vec{b})(-\vec{a} + \vec{b}) = 0$ or $4\vec{a}^2 - 4\vec{a}.\vec{b} + \vec{a}.\vec{b} - \vec{b}^2 = 0$

or $4\vec{a}^2 - \vec{b}^2 - 3\vec{a}.\vec{b} = 0$ ....(2)

now $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s}) = \frac{1}{3}(5\vec{a} - 3\vec{b} - 4\vec{a} - \vec{b} - \vec{a} + \vec{b}) = -\vec{b}$

$\vec{y} = \frac{1}{5}(\vec{r} + \vec{s}) = \frac{1}{5}(-5\vec{a}) = -\vec{a}$

Angle between $\vec{x}$ and $\vec{y}$ i.e. $\cos\theta = \dfrac{\vec{x}.\vec{y}}{|\vec{x}||\vec{y}|} = \dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}$ ....(3)

From (1) and (2) $|\vec{a}| = \sqrt{\dfrac{25}{19}}\sqrt{\vec{a}.\vec{b}}$ and $|\vec{b}| = \sqrt{\dfrac{43}{19}}\sqrt{\vec{a}.\vec{b}}$

$\therefore \quad |\vec{a}||\vec{b}| = \dfrac{\sqrt{25\times 43}}{19}.\vec{a}.\vec{b}$; $\quad \therefore \quad \dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} = \dfrac{19}{5\sqrt{43}} \quad \Rightarrow \quad \theta = \cos^{-1}\dfrac{19}{5\sqrt{43}}$

7. If the vector product of a constant vector $\vec{OA}$ with a variable vector $\vec{OB}$ in a fixed plane OAB be a constant vector, then locus of B is :

(A) a straight line perpendicular to $\vec{OA}$ (B) a circle with centre O radius equal to $\left|\vec{OA}\right|$

(C) a straight line parallel to $\vec{OA}$ (D) None of these

यदि एक नियत सदिश $\vec{OA}$ का नियत तल OAB में एक चर सदिश $\vec{OB}$ के साथ सदिश गुणन एक नियत सदिश है तो B का बिन्दुपथ है :

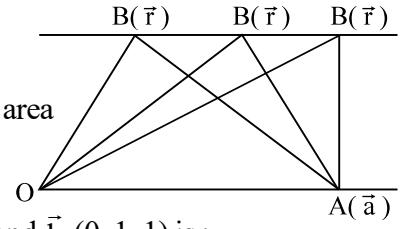
(A) एक सरल रेखा जो $\vec{OA}$ के लम्बवत् है (B) एक वृत जिसका केन्द्र O तथा त्रिज्या $\left|\vec{OA}\right|$ है

(C) $\vec{OA}$ के समान्तर रेखा है (D) इनमें से कोई नहीं

Ans. C

Sol. $|\vec{a}\times\vec{r}| = |\vec{c}|$

Triangles on the same base and between the same parallel will have the same area



8. The number of vectors of unit length perpendicular to vectors $\vec{a} = (1, 1, 0)$ and $\vec{b}$ $(0, 1, 1)$ is :

सदिशों $\vec{a} = (1, 1, 0)$ तथा $\vec{b}$ $(0, 1, 1)$ के लम्बवत् इकाई सदिशों की संख्या है :

(A) 1 (B) 2 (C) 3 (D) ∞

Ans. B

Sol. Two collinear vector always denotes a plane

9. Let O be an interior point of ΔABC such that $2\overrightarrow{OA} + 5\overrightarrow{OB} + 10\overrightarrow{OC} = \vec{0}$. If the ratio of the area of ΔABC to the area of ΔAOC is t, where 'O' is the origin. Find [t] : (Where [ ] denotes greatest integer function)

माना O त्रिभुज ABC के अन्तः में स्थित एक बिन्दु इस प्रकार है कि $2\overrightarrow{OA} + 5\overrightarrow{OB} + 10\overrightarrow{OC} = \vec{0}$ यदि क्षेत्रफल (ΔABC) का

क्षेत्रफल (ΔAOC) से अनुपात t है, जहाँ 'O' मूल बिन्दु है। [t] का ज्ञात कीजिए। (जहाँ [ ] महत्तम पूर्णांक फलन है)

Ans. 3

Sol. $t = \frac{32}{5} \Rightarrow [t] = 6$

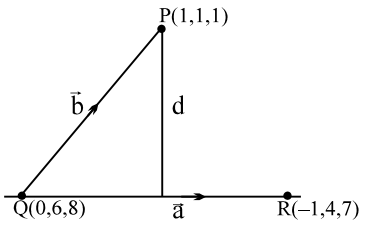
10. If the distance from the point P(1, 1, 1) to the line passing through the points Q(0, 6, 8) and R(–1, 4, 7) is expressed in the form $\sqrt{\frac{p}{q}}$ where p and q are coprime, then find the value of $\frac{(p+q)(p+q-1)}{2}$.

यदि बिन्दु P(1, 1, 1) की बिन्दुओं Q(0, 6, 8) तथा R(–1, 4, 7) को मिलाने वाली रेखा से दूरी $\sqrt{\frac{p}{q}}$ के रूप में व्यक्त होती है जहाँ p तथा q सापेक्षतः अभाज्य है, $\frac{(p+q)(p+q-1)}{2}$ का मान ज्ञात कीजिए।

Ans. 4950

Sol. $\vec{a} = -\hat{i} - 2\hat{j} - \hat{k}$

$\vec{b} = \hat{i} - 5\hat{j} - 7\hat{k}$



$\left|\vec{d}\right| = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}|}$

$|\vec{a}| = \sqrt{6}$

$|\vec{a}\times\vec{b}|^2 = \vec{a}^2\vec{b}^2 - (\vec{a}\cdot\vec{b})^2 = (6)(75) - (-1 + 10 + 7)^2 = 450 - 256 = 194$

$|\vec{a}\times\vec{b}| = \sqrt{194}$

$\therefore\ d = \sqrt{\frac{194}{6}} = \sqrt{\frac{97}{3}}$

$\therefore\ p + q = 100 \qquad \Rightarrow \qquad \frac{(p+q)(p+q-1)}{2} = \frac{100\times 99}{2} = 4950$

11. Given $f^2(x) + g^2(x) + h^2(x) \le 9$ and $U(x) = 3f(x) + 4g(x) + 10h(x)$, where f (x), g (x) and h (x) are continuous $\forall\ x \in R$. If maximum value of U(x) is $\sqrt{N}$, then find N.

दिया गया है $f^2(x) + g^2(x) + h^2(x) \le 9$ तथा $U(x) = 3f(x) + 4g(x) + 10h(x)$, जहाँ f (x), g (x) तथा h (x) सतत् है सभी $x \in R$ के लिये। यदि U(x) का अधिकतम मान $\sqrt{N}$ है तो N का मान ज्ञात कीजिए।

Ans. 1125

Sol. Let $\vec{V}_1 = 3\hat{i} + 4\hat{j} + 10\hat{k}$ and $\vec{V}_2 = f(x)\hat{i} + g(x)\hat{j} + h(x)\hat{k}$

$U(x) = \vec{V}_1 \cdot \vec{V}_2 = |\vec{V}_1||\vec{V}_2|\cos\theta$

$\leq |\vec{V}_1||\vec{V}_2|$

$= \sqrt{9+16+100}\sqrt{f^2+g^2+h^2} \leq 3\sqrt{125} = \sqrt{1125} \Rightarrow N = 1125 =$ Ans.

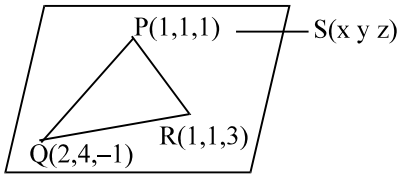
1. Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}$, $\vec{q}$ and $\vec{r}$ denotes the position vector of three non-collinear points then the equation of the plane containing these points is :

माना तीन सदिश $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ तथा $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$ है। यदि $\vec{p}$, $\vec{q}$ तथा $\vec{r}$ तीन विषमरेखीय बिन्दुओं के स्थिति सदिश है तो तीनों बिन्दु द्वारा बनाए तल का समीकरण होगा :

(A) $2x - 3y + 1 = 0$ (B) $x - 3y + 2z = 0$ (C) $3x - y + z - 3 = 0$ (D) $3x - y - 2 = 0$

Ans. D

Sol. Equation of plane $\begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & 3 & -2 \\ 0 & 0 & 2 \end{vmatrix} = 0$



$\Rightarrow \quad 2(3x - 3 - y + 1) = 0 \quad \Rightarrow \quad 3x - y = 2$

2. The intercept made by the plane $\vec{r}.\vec{n} = q$ on the x-axis is :

समतल $\vec{r}.\vec{n} = q$ द्वारा x-अक्ष पर काटा गया अन्तःखण्ड है :

(A) $\frac{q}{\hat{i}.\vec{n}}$ (B) $\frac{\hat{i}.\vec{n}}{q}$ (C) $\left(\hat{i}.\vec{n}\right)q$ (D) $\frac{q}{|\vec{n}|}$

Ans. A

Sol. x intercept is say $x_1 \Rightarrow$ plane passes through $x_1\hat{i}$

$\therefore \quad x_1\hat{i}.\vec{n} = q \Rightarrow x_1 = \frac{q}{\hat{i}.\vec{n}}$

3. If the distance between the planes $8x + 12y - 14z = 2$ and $4x + 6y - 7z = 2$ can be expressed in the form $\frac{1}{\sqrt{N}}$ where N is natural then the value of $\frac{N(N+1)}{2}$ is :

तालों $8x + 12y - 14z = 2$ एवं $4x + 6y - 7z = 2$ के मध्य दूरी $\frac{1}{\sqrt{N}}$ के रूप में व्यक्त की जाती है, जहाँ N एक प्राकृतिक संख्या है, तो $\frac{N(N+1)}{2}$ का मान है :

(A) 4950 (B) 5050 (C) 5150 (D) 5151

Ans. D

Sol. $P_1 = 4x + 6y - 7z - 1 = 0$; $P_2 = 4x + 6y - 7z - 2 = 0$

$d = \frac{1}{\sqrt{16+36+49}} = \frac{1}{\sqrt{101}}$

Hence, $\frac{101\times102}{2} = 5151$ $\Rightarrow$ (D)

4. A plane passes through the point P(4, 0, 0) and Q(0, 0, 4) and is parallel to the y-axis. The distance of the plane from the origin is :

एक तल बिन्दु P(4, 0, 0) तथा Q(0, 0, 4) से गुजरता है तथा y-अक्ष के समानान्तर है। तल की मूल बिन्दु से दूरी है :

(A) 2 (B) 4 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Ans. D

Sol. x and z intercept of the plane is 4 and it is parallel to y-axis, hence equation of the plane is $x + z = 4$.

Its distance from (0, 0, 0) is $2\sqrt{2}$

5. If from the point P (f, g, h) perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is :

यदि बिन्दु P (f, g, h) से yz तथा zx तलों पर डाले गए लम्ब PL तथा PM क्रमशः है तो OLM तल का समीकरण है :

(A) $\frac{x}{f}+\frac{y}{g}-\frac{z}{h}=0$ (B) $\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$ (C) $\frac{x}{f}-\frac{y}{g}+\frac{z}{h}=0$ (D) $-\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$

Ans. A

Sol. coordinate of L (0, g, h) and M (f, 0, h). Now find the equation of OLM

6. If the plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(k)$ with x-axis, then k is equal to :

यदि तल $2x - 3y + 6z - 11 = 0$ x-अक्ष के साथ $\sin^{-1}(k)$ कोण बनाता है तो k बराबर होगा :

(A) $\frac{\sqrt{3}}{2}$ (B) 2/7 (C) $\frac{\sqrt{2}}{3}$ (D) 1

Ans. B

Sol. $\vec{n} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ and $\vec{V} = \hat{i}$; now $\sin\theta = \frac{\vec{V}.\vec{n}}{|\vec{n}|} = \frac{2}{7}$ $\Rightarrow$ $K = \frac{2}{7}$

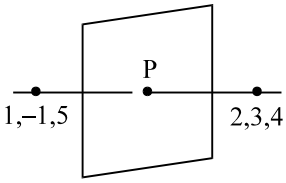
7. The plane XOZ divides the join of (1, –1, 5) and (2, 3, 4) in the ratio $\lambda : 1$, then λ is :

तल XOZ बिन्दु (1, –1, 5) तथा (2, 3, 4) को मिलाने वाली रेखा को $\lambda : 1$ अनुपात में विभाजित करता है, तो λ =

(A) – 3 (B) – 1/3 (C) 3 (D) 1/3

Ans. D

Sol. y coordinate of P is zero $\Rightarrow 0 = \frac{3\lambda + (-1)}{\lambda + 1} \Rightarrow \lambda = \frac{1}{3}$



8. Equation of the line which passes through the point with p. v. (2, 1, 0) and perpendicular to the plane containing the vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$ is : (where t is a parameter)

उस रेखा का समीकरण ज्ञात कीजिये जो (2, 1, 0) बिन्दु से गुजरती है तथा सदिशों $\hat{i} + \hat{j}$ तथा $\hat{j} + \hat{k}$ द्वारा बनाये गये समतल के लम्बवत् है : (जहाँ t प्राचाल है)

(A) $\vec{r}$ = (2, 1, 0) + t (1, –1, 1) (B) $\vec{r}$ = (2, 1, 0) + t (–1, 1, 1)

(C) $\vec{r}$ = (2, 1, 0) + t (1, 1, –1) (D) $\vec{r}$ = (2, 1, 0) + t (1, 1, 1)

Ans. A

Sol. $\vec{r} = 2\hat{i} + \hat{j} + 0\hat{k} + t\ (\hat{i} + \hat{j}) \times (\hat{j} + \hat{k}) = (2, 1, 0) + t\ (\hat{k} - \hat{j} + \hat{i}) = (2, 1, 0) + t\ (1, -1, 1) \Rightarrow$ (A)

9. Which of the following planes are parallel but not identical?

$P_1 : 4x - 2y + 6z = 3$

$P_2 : 4x - 2y - 2z = 6$

$P_3 : -6x + 3y - 9z = 5$

$P_4 : 2x - y - z = 3$

(A) P_2 and P_3 (B) P_2 and P_4 (C) P_1 and P_3 (D) P_1 and P_4

निम्न में से कौनसे तल समानान्तर है किन्तु संपाती नही है ?

$P_1 : 4x - 2y + 6z = 3$

$P_2 : 4x - 2y - 2z = 6$

$P_3 : -6x + 3y - 9z = 5$

$P_4 : 2x - y - z = 3$

(A) P_2 तथा P_3 (B) P_2 तथा P_4 (C) P_1 तथा P_3 (D) P_1 तथा P_4

Ans. C

Sol. In A, $-2/3 = -2/3 \neq 2/9$

B, $2 = 2 = 2 = 2$ identical

C, $-2/3 = -2/3 = -2/3 \neq 3/5 \Rightarrow$ (C) is correct

D, $2 = 2 \neq -6$

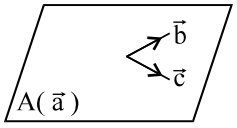
10. Vector equation of the plane $\vec{r}=\hat{i}-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})+\mu(\hat{i}-2\hat{j}+3\hat{k})$ in the scalar dot product form is :

सदिश समीकरण $\vec{r}=\hat{i}-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})+\mu(\hat{i}-2\hat{j}+3\hat{k})$ द्वारा दर्शाये गये तल को अदिश गुणनफल रूप में व्यक्त कीजिए :

(A) $\vec{r}.(5\hat{i}-2\hat{j}+3\hat{k})=7$ (B) $\vec{r}.(5\hat{i}+2\hat{j}-3\hat{k})=7$ (C) $\vec{r}.(5\hat{i}-2\hat{j}-3\hat{k})=7$ (D) $\vec{r}.(5\hat{i}+2\hat{j}+3\hat{k})=7$

Ans. C

Sol. plane through $\vec{a}$ and $\parallel$ to two non collinear vector $(\vec{r}-\vec{a}).(\vec{b}\times\vec{c})=0$



(Take dot with $\vec{b}\times\vec{c}$ both sides]

11. The vector equations of the two lines L_1 and L_2 are given by

$L_1 : \vec{r} = 2\hat{i}+9\hat{j}+13\hat{k}+\lambda(\hat{i}+2\hat{j}+3\hat{k})$; $L_2 : \vec{r} = -3\hat{i}+7\hat{j}+p\hat{k}+\mu(-\hat{i}+2\hat{j}-3\hat{k})$

then the lines L_1 and L_2 are :

(A) skew lines for all $p \in R$

(B) intersecting for all $p \in R$ and the point of intersection is $(-1, 3, 4)$

(C) intersecting lines for $p = -2$

(D) intersecting for all real $p \in R$

दो रेखाओं L_1 तथा L_2 का सदिश समीकरण

$L_1 : \vec{r} = 2\hat{i}+9\hat{j}+13\hat{k}+\lambda(\hat{i}+2\hat{j}+3\hat{k})$; $L_2 : \vec{r} = -3\hat{i}+7\hat{j}+p\hat{k}+\mu(-\hat{i}+2\hat{j}-3\hat{k})$

है तो L_1 तथा L_2 है :

(A) सभी $p \in R$ के लिये विसमतलीय

(B) सभी $p \in R$ के लिये प्रतिच्छेदी तथा उनका प्रतिच्छेदन बिन्दु $(-1, 3, 4)$ है

(C) $p = -2$ के लिये प्रतिच्छेदी

(D) सभी $p \in R$ के लिये प्रतिच्छेदी

Ans. C

Sol. Intersecting if $\begin{vmatrix} 5 & 2 & 13-p \\ 1 & 2 & 3 \\ -1 & 2 & -3 \end{vmatrix} = \begin{vmatrix} 5 & 2 & 13-p \\ 0 & 4 & 0 \\ -1 & 2 & -3 \end{vmatrix}$ or $-4(-15+13-p)=0$

$p = -2$

Alternatively: $(\lambda + 2) = -(\mu + 3)$ _____(1)

$2\lambda + 9 = 2\mu + 7$ _____(2)

$3\lambda + 13 = p - 3\mu$ _____(3)

from(1) $\mu = -(\lambda + 5)$

put in (2) $2\lambda + 9 = -2(\lambda + 5) + 7 \Rightarrow \lambda = -3$

now from (3) $-9 + 13 = p + 6 \Rightarrow p = -2$ Ans.

12. Consider the plane $(x, y, z) = (0, 1, 1) + \lambda(1, -1, 1) + \mu(2, -1, 0)$. The distance of this plane from the origin is :

माना समतल $(x, y, z) = (0, 1, 1) + \lambda(1, -1, 1) + \mu(2, -1, 0)$ है। मूल बिन्दु से इस तल की दूरी है :

(A) 1/3 (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{2}{\sqrt{3}}$

Ans. C

Sol. $\vec{r} = \vec{a} + \lambda\vec{b} + \mu\vec{c}$

taking dot with $\vec{b} \times \vec{c}$

$[\vec{r}\ \vec{b}\ \vec{c}] = [\vec{a}\ \vec{b}\ \vec{c}]$ where $\vec{a} = (0,1,1)$; $\vec{b} = (1,-1,1)$ and $\vec{c} = (2,-1,0)$

$$[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} 0 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & -1 & 0 \end{vmatrix} = 0 - (0 - 2) + 1(-1 + 2) = 3$$

and $$[\vec{r}\ \vec{b}\ \vec{c}] = \begin{vmatrix} x & y & z \\ 1 & -1 & 1 \\ 2 & -1 & 0 \end{vmatrix} = x(0 + 1) - y(0 - 2) + z(-1 + 2) = x + 2y + z$$

hence equation of plane is $x + 2y + z = 3$; $\therefore\ p = \left|\frac{-3}{\sqrt{6}}\right| = \sqrt{\frac{3}{2}}$ Ans.

13. Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$. then (α, β) equals : [AIEEE-2009]

रेखा $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ समतल $x + 3y - \alpha z + \beta = 0$ में स्थित है तो (α, β) बराबर होगा : [AIEEE-2009]

(A) (–6, 7) (B) (5, –15) (C) (–5, 5) (D) (6, –17)

Ans. A

Sol. Line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ _____(1)

Plane $x + 3y - \alpha z + \beta = 0$ _____(2)

Point (2, 1, –2) put in (2)

$2 + 3 + 2\alpha + \beta = 0$

$\Rightarrow \quad 2\alpha + \beta = -5$

Now $a_1a_2 + b_1b_2 + c_1c_2 = 0$

$3 - 15 - 2\alpha = 0$

$-12 - 2\alpha = 0$

$\alpha = -6$

$-12 + \beta = -5$
$\beta = 7$
$\alpha = -6, \beta = 7$

14. The projections of a vector on the three coordinate axis are 6, –3, 2 respectively. The direction cosines of the vector are : [AIEEE-2009]

तीन निर्देशी अक्षों पर एक सदिश के प्रक्षेप क्रमशः 6, –3, 2 हैं। सदिश की दिक्कोज्याऐं है : [AIEEE-2009]

(A) $\frac{6}{5}, \frac{-3}{5}, \frac{2}{5}$ (B) $\frac{6}{7}, \frac{-3}{7}, \frac{2}{7}$ (C) $\frac{-6}{7}, \frac{-3}{7}, \frac{2}{7}$ (D) 6, –3, 2

Ans. B

Sol. Proj. of a vector $(\vec{r})$ on x–axis $= |\vec{r}|\,\ell$

on y–axis $= |\vec{r}|\,m$

on z–axis $= |\vec{r}|\,n$

$6 = 7\ell, \Rightarrow \ell = \frac{6}{7}$ similarly $m = -\frac{3}{7}, n = \frac{2}{7}$

15. If the angle between the line $x = \frac{y-1}{2\lambda} = \frac{z-3}{\lambda}$ and the plane $x + 2y + 3z = 4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals : [AIEEE-2011]

यदि रेखा $x = \frac{y-1}{2\lambda} = \frac{z-3}{\lambda}$ तथा समतल $x + 2y + 3z = 4$ के बीच का कोण $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$ है, तो λ का मान है : [AIEEE-2011]

(A) $\frac{2}{5}$ (B) $\frac{5}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Ans. D

Sol. $\frac{x}{1} = \frac{y-1}{2} = \frac{z-3}{\lambda}$ equation of line

equation of plane $x + 2y + 3z = 4$

$\sin\theta = \frac{1+4+3\lambda}{\sqrt{14}\sqrt{1+4+\lambda^2}}$

$\Rightarrow \lambda = \frac{2}{3}$

16. The distance of the point (1, 0, 2) from the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=16$, i : [JEE(Main)-2015]

रेखा $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ तथा तल $x-y+z=16$ के प्रतिच्छेद बिन्दु का (1, 0, 2) से दूरी है : [JEE(Main)-2015]

(A) $2\sqrt{14}$ (B) 8 (C) $3\sqrt{21}$ (D) 13

Ans. D

Sol. $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}=\lambda$

POI $(2+3\lambda, 4\lambda-1, 12\lambda+2)$

$(2+3\lambda)-(4\lambda-1)+12\lambda+2=16$

$\lambda=1$

So Point of Intersection (5, 3, 14)

Distance $=\sqrt{16+9+144}=13$

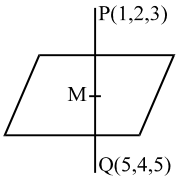
17. The equation of the plane which has the property that the point Q (5, 4, 5) is the reflection of point P (1, 2, 3) through that plane, is $ax+by+cz=d$ where $a, b, c, d \in N$. Find the least value of $(a+b+c+d)$.

एक तल का समीकरण जिसका गुणधर्म हैं कि Q (5, 4, 5) बिन्दु P (1, 2, 3) का तल में प्रतिबिम्ब है, $ax+by+cz=d$ है जहाँ $a, b, c, d \in N$ है। $(a+b+c+d)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 17

Sol. $M \equiv (3, 3, 4)$

and normal vector of the plane is parallel to $\overrightarrow{PQ}$



Hence $\vec{n}=4\hat{i}+2\hat{j}+2\hat{k}=2\left(2\hat{i}+\hat{j}+\hat{k}\right)$

$\therefore$ Equation of plane passing through the midpoint M of PQ, is

$2(x-3)+1(y-3)+1(z-4)=0$

$2x+y+z=13 \equiv ax+by+cz=d$

$\Rightarrow$ $a=2, b=1, c=1, d=13$ $\Rightarrow$ $a+b+c+d=17$ Ans.

1. The value of 'a' for which the lines $\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ and $\frac{x-a}{-1}=\frac{y-7}{2}=\frac{z+2}{-3}$ intersect, is :

'a' के किन मानों के लिये रेखायें $\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ तथा $\frac{x-a}{-1}=\frac{y-7}{2}=\frac{z+2}{-3}$ प्रतिच्छेद करती है :

(A) – 5 (B) – 2 (C) 5 (D) – 3

Ans. D

Sol. $\begin{vmatrix} 2-a & 9-7 & 13-(-2) \\ 1 & 2 & 3 \\ -1 & 2 & -3 \end{vmatrix}=0$

2. Given A (1, –1, 0) ; B(3, 1, 2); C(2, –2, 4) and D(–1, 1, –1) which of the following points neither lie on AB nor on CD?

दिया गया है A (1, –1, 0) ; B(3, 1, 2) ; C(2, –2, 4) तथा D(–1, 1, –1) निम्न में से कौनसे बिन्दु कभी भी AB अथवा CD पर स्थित नही होगें?

(A) (2, 2, 4) (B) (2, –2, 4) (C) (2, 0,1) (D) (0, –2, –1)

Ans. A

Sol. Write the lines AB and CD in symmetrical form and verify

3. For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect?

(A) it lies in the plane $x-2y+z=0$ (B) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) it passes through (2, 3, 5) (D) it is parallel to the plane $x-2y+z-6=0$

रेखा $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ के लिये, निम्न में से कौनसे कथन असत्य है?

(A) यह तल $x-2y+z=0$ पर स्थित है (B) यह रेखा $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ के समान है

(C) यह (2, 3, 5) से गुजरती है (D) यह तल $x-2y+z-6=0$ के समानान्तर है

Ans. C

Sol. On (1, 2, 3) satisfies the plane $x - 2y + z = 0$ and also $(\hat{i} + 2\hat{j} + 3\hat{k}).(\hat{i} - 2\hat{j} + \hat{k}) = 0 \Rightarrow$(A)

Since the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ both satisfy $t + 1, 2t + 2$ and $3t + 3$

Hence both are same $\Rightarrow$ (B). Given line is obviously $||$ to the plane $x - 2y + z = 6 \Rightarrow$ (D)

4. Given planes

$P_1 : cy + bz = x$

$P_2 : az + cx = y$

$P_3 : bx + ay = z$

P_1, P_2 and P_3 pass through one line, if :

दिया गया तल

$P_1 : cy + bz = x$

$P_2 : az + cx = y$

$P_3 : bx + ay = z$

P_1, P_2 तथा P_3 एक ही रेखा से होकर गुजरते है, यदि :

(A) $a^2 + b^2 + c^2 = ab + bc + ca$ (B) $a^2 + b^2 + c^2 + 2abc = 1$

(C) $a^2 + b^2 + c^2 = 1$ (D) $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + 2abc = 1$

Ans. B

Sol. Infinite solution $\Rightarrow \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0 \Rightarrow \vec{n}_1 \times \vec{n}_2 \cdot \vec{n}_3$

note that 3 such planes can meet only at one point i.e. (0, 0, 0) or they may have the same line of intersection i.e. at infinite solution.

5. यह रेखा $\frac{x - x_1}{0} = \frac{y - y_1}{1} = \frac{z - z_1}{2}$ है :

(A) parallel to x-axis (B) perpendicular to x-axis

(C) perpendicular to YOZ plane (D) parallel to y-axis

The line $\frac{x - x_1}{0} = \frac{y - y_1}{1} = \frac{z - z_1}{2}$ is :

(A) x-अक्ष के सामानान्तर (B) x-अक्ष के लम्बवत्

(C) YOZ तल के लम्बवत् (D) y-अक्ष के समानान्तर

Ans. B

Sol. $L_1 : \frac{x - x_1}{0} = \frac{y - y_1}{1} = \frac{z - z_1}{2}$ x–axis = (1, 0, 0)

L_1 perpendicular to x–axis.

6. The lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar if :

रेखायें $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ तथा $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ समतलीय है यदि :

(A) k = 0 or – 1 (B) k = 1 or – 1 (C) k = 0 or – 3 (D) k = 3 or – 3

Ans. C

Sol. The given lines are coplanar if

$$\begin{vmatrix} 2-1 & 3-4 & 4-5 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 1-k \\ k & k+2 & 1+k \end{vmatrix}$$

or if $2(1+k)-(k+2)(1-k)=0$ if $k^2+3k=0$
or if k = 0 or – 3

7. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in N$ and $a > 3$ then (a + b) equals :

(A) 23 (B) 95 (C) 27 (D) None

रेखा, जो (x, y, z) द्वारा दर्शाये सभी बिन्दुओं से गुजरती है जिनके लिये $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ तल $2x - 3y + 4z = 163$ को बिन्दु P पर प्रतिच्छेद करती है तथा YZ को Q पर काटते है। यदि $PQ = a\sqrt{b}$ जहाँ $a, b \in N$ तथा $a > 3$ तो (a + b) बराबर होगा :

(A) 23 (B) 95 (C) 27 (D) कोई नहीं

Ans. A

Sol. Equation of the line is

$$\frac{x-2}{1}=\frac{y+2}{-3}=\frac{z-5}{2}=\lambda \quad ....(1)$$

hence any point on the line (1) can be taken as

$\Rightarrow$ $x = \lambda + 2$
$y = -(3\lambda + 2)$
$z = (2\lambda + 5)$

for some λ point lies on the plane

$2x - 3y + 4z = 163$ ....(2)
$2(\lambda + 2) + 3(3\lambda + 2) + 4(2\lambda + 5) = 163$

This gives $19\lambda = 133 \Rightarrow \lambda = 7$

Hence, $P \equiv (9, -23, 19)$

Also (1) intersect YZ plane i.e. $x = 0 \Rightarrow \lambda + 2 = 0$, hence $\lambda = -2$

$\therefore$ Q(0, 4, 1)

So, $PQ = \sqrt{9^2 + 27^2 + 18^2} = 9\sqrt{1 + 3^2 + 2^2} = 9\sqrt{14}$

$\therefore$ a = 9 and b = 14

Hence, a + b = 23

8. Let L_1 be the line $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ and let L_2 be the line $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$.

Let Π be the plane which contains the line L_1 and is parallel to L_2. The distance of the plane Π from the origin is :

(A) 1/7 (B) $\sqrt{\frac{2}{7}}$ (C) $\sqrt{6}$ (D) None

माना L_1 एक रेखा है $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ तथा L_2 एक रेखा $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$ है। यदि Π एक तल है जो रेखा L_1 से गुजरता है तथा L_2 के समान्तर है। तल Π की मूल बिन्दु से दूरी है :

(A) 1/7 (B) $\sqrt{\frac{2}{7}}$ (C) $\sqrt{6}$ (D) कोई नहीं

Ans. B

Sol. Equation of the plane containing L_1

$A(x - 2) + B(y - 1) + C(z + 1) = 0$

where $A + 2C = 0$ also, $A + B - C = 0$

$\Rightarrow$ $A = -2C$, $B = 3C$, $C = C$

plane is $-2(x - 2) + 3(y - 1) + z + 1 = 0$

$-2x + 3y + z + 4 - 3 + 1 = 0$

$2x - 3y - z - 2 = 0$

Hence $p = \left|\frac{-2}{\sqrt{14}}\right| = \sqrt{\frac{2}{7}}$ Ans.

9. The value of m for which straight line $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ is parallel to the plane $2x - y + mz - 2 = 0$ is :

m के मान जिनके लिये सरल रेखा $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ समतल $2x - y + mz - 2 = 0$ के समानान्तर है :

(A) –2 (B) 8 (C) – 18 (D) 11

Ans. A

Sol. Vector $\left((3\hat{i}-2\hat{j}+\hat{k})\times(4\hat{i}-3\hat{j}+4\hat{k})\right)$ is perpendicular to $2\hat{i}-\hat{j}+m\hat{k} \Rightarrow \begin{vmatrix} 3 & -2 & 1 \\ 4 & -3 & 4 \\ 2 & -1 & m \end{vmatrix} = 0$

10. The distance of the point (–1, –5, – 10) from the point of intersection of the line $\frac{x-2}{2}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=5$ is :

बिन्दु की रेखा (–1, –5, – 10) की रेखा $\frac{x-2}{2}=\frac{y+1}{4}=\frac{z-2}{12}$ एवं तल $x-y+z=5$ के प्रतिच्छेदन बिन्दु से दूरी है :

(A) $2\sqrt{11}$ (B) $\sqrt{126}$ (C) 13 (D) 14

Ans. C

Sol. Any point on $\frac{x-2}{2}=\frac{y+1}{4}=\frac{z-2}{12}$ can be (2r + 2, 4r – 1, 12r + 2)

which lies on $x-y+z=5$

$\therefore$ $(2r+2)-(4r-1)+12r+2=5$

$\Rightarrow$ $r=0$

$\therefore$ Point on the plane $\equiv (2,-1,2)$

Distance between (2, – 1, 2) and (–1, –5, – 10) = $\sqrt{(2+1)^2+(-1+5)^2+(2+10)^2}$ = 13 Ans.

11. $P(\vec{p})$ and $Q(\vec{q})$ are the position vectors of two fixed points and $R(\vec{r})$ is the position vector of a variable point. If R moves such that $(\vec{r}-\vec{p})\times(\vec{r}-\vec{q})=0$ then the locus of R is :

(A) a plane containing the origin 'O' and parallel to two non collinear vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$

(B) the surface of a sphere described on PQ as its diameter.

(C) a line passing through the points P and Q

(D) a set of lines parallel to the line PQ

$P(\vec{p})$ तथा $Q(\vec{q})$ दो नियत बिन्दुओं के स्थिति सदिश है तथा $R(\vec{r})$ एक चर बिन्दु का स्थिति सदिश है। यदि R इस प्रकार गमन करता है कि $(\vec{r}-\vec{p})\times(\vec{r}-\vec{q})=0$ तो R का बिन्दुपथ है :

(A) मूल बिन्दु 'O' से गुजरने वाला तल जो सदिशों $\overrightarrow{OP}$ तथा $\overrightarrow{OQ}$ के समानान्तर है

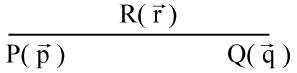
(B) PQ व्यास पर बनाये गये गोले का सतह

(C) एक रेखा जो बिन्दुओं P तथा Q से गुजरती है

(D) रेखा PQ के समानान्तर रेखाओं का समुच्चय

Ans. C

Sol. Obviously (C) ; $R(\vec{r})$ moves on PQ



12. Let L be the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$. If L makes an angle α with the positive x-axis, then $\cos\alpha$ equals : [AIEEE-2007]

माना L रेखा समतल $2x + 3y + z = 1$ तथा $x + 3y + 2z = 2$ को प्रतिच्छेद करती है। यदि L धनात्मक x-अक्ष के साथ α कोण बनाती है तो $\cos\alpha$ बराबर है : [AIEEE-2007]

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{\sqrt{2}}$

Ans. A

Sol. If direction cosines of L be ℓ, m, n then

$2\ell + 3m + n = 0$

$\ell + 3m + 2n = 0$

Solving, we get, $\frac{\ell}{3} = \frac{m}{-3} = \frac{n}{3}$

$\therefore \ \ell : m : n = \frac{1}{\sqrt{3}} : -\frac{1}{\sqrt{3}} : \frac{1}{\sqrt{3}} \Rightarrow \cos\alpha = \frac{1}{\sqrt{3}}$

13. Statemenrt-1: The point A(3, 1, 6) is the mirror image of the point B(1, 3, 4) in the plane $x - y + z = 5$.

Statement-2: The plane $x - y + z = 5$ bisects the line segment joining A(3, 1, 6) and B(1, 3, 4). [AIEEE-2010]

(A) Statemenrt-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

प्रकथन-1: तल $x - y + z = 5$ में बिन्दु B(1, 3, 4) को दर्पण प्रतिबिम्ब बिन्दु A(3, 1, 6) है।

प्रकथन-2: तल $x - y + z = 5$, A(3, 1, 6) तथा B(1, 3, 4) को मिलाने वाले रेखाखण्ड को समद्विभाजित करता है। [AIEEE-2010]

(A) प्रकथन-1 सत्य है, प्रकथन -2 सत्य है। प्रकथन -2 प्रकथन -1 की सही व्याख्या है

(B) प्रकथन-1 सत्य है, प्रकथन -2 सत्य है। प्रकथन -2 प्रकथन -1 की सही व्याख्या नहीं है

(C) प्रकथन-1 सत्य है, प्रकथन -2 असत्य है

(D) प्रकथन -1 असत्य है, प्रकथन -2 सत्य है

Ans. B

Sol. Mirror image of B(1, 3, 4) in plane $x-y+z = 5$

$\frac{x-1}{1} = \frac{y-3}{-1} = \frac{z-4}{1} = -2\frac{(1-3+4-5)}{1+1+1} = 2$

$\Rightarrow x = 3, y = 1, z = 6$

$\therefore$ mirror image of B (1, 3, 4)
is A (3, 1, 6)
statement-1 is correct
statement-2 is true but it is not the correct explanation.

14. Statement-1 : The point A(1, 0, 7) is the mirror image of the point B(1, 6, 3) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$.

Statment-2: The line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ bisects the line segment joining A(1, 0, 7) and B(1, 6, 3).

[AIEEE-2011]

(A) Statement-1 is true, Statment-2 is false

(B) Statement-1 is false, Statement-2 is true

(C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(D) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

कथन-1 : रेखा $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ में बिन्दु B(1, 6, 3) का दर्पण प्रतिबिम्ब बिन्दु A(1, 0, 7) है।

कथन-2: रेखा $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ बिन्दुओं A(1, 0, 7) तथा B(1, 6, 3) को मिलाने वाले रेखाखण्ड का समद्विभाजन करती है।

[AIEEE-2011]

(A) कथन -1 सत्य है, कथन -2 असत्य है

(B) कथन-1 असत्य है, कथन-2 सत्य है

(C) कथन-1 सत्य है, कथन-2 सत्य है। कथन-2, कथन -1 की सही व्याख्या है

(D) कथन-1 सत्य है, कथन-2 सत्य है। कथन -2 कथन -1 की सही व्याख्या नहीं है

Ans. D

Sol. $1(1-1) + 2(0-6) + 3(7-3)$
$= 0 - 12 + 12 = 0$
mid point AB (1, 3, 5)
lies on $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$

15. An equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is :
[AIEEE-2012]

समतल $x - 2y + 2z - 5 = 0$ के समांतर तथा मूल बिन्दु से इकाई दूरी पर एक समतल का समीकरण है : [AIEEE-2012]

(A) $x-2y+2z+5=0$ (B) $x - 2y + 2z - 3 = 0$ (C) $x - 2y + 2z+1 = 0$ (D) $x - 2y + 2z - 1 = 0$

Ans. B

Sol. Equation of plane parallel to
$x - 2y + 2z - 5 = 0$
is $x - 2y + 2z = k$

or $\frac{x}{3} - \frac{2}{3}y + \frac{2}{3}z = \frac{K}{3}$

$\left|\frac{K}{3}\right| = 1$

$\Rightarrow K = \pm 3$

$\therefore$ Equation of required plane is
$x - 2y + 2z \pm 3 = 0$

16. The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line. [JEE(Main)-2014]

समतल $2x - y + z + 3 = 0$ में रेखा $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ के प्रतिबिम्ब वाली रेखा है। [JEE(Main)-2014]

(A) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$ (B) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$

(C) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$ (D) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

Ans. C

Sol. Any point on the given $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ will be $(3t+1, t+3, -5t+4)$ its image in the plane

$2x - y + z + 3 = 0$ is given by $\frac{x-(3t+1)}{2} = \frac{y-(t+3)}{-1} = \frac{z-(-5t+4)}{1} = -2\frac{(6t+2-t-3-5t+4+3)}{6} = -2$

$\Rightarrow x = 3t - 3, y = t + 5, z = 2 - 5t$ which is the line $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$ So (C) is correct.

1. For non-zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\left|\vec{a} \times \vec{b}.\vec{c}\right| = |\vec{a}|\,|\vec{b}|\,|\vec{c}|$ holds if and only if :

अशून्य सदिशों $\vec{a}$, $\vec{b}$, $\vec{c}$ के लिये, $\left|\vec{a} \times \vec{b}.\vec{c}\right| = |\vec{a}|\,|\vec{b}|\,|\vec{c}|$ सही है यदि और केवल यदि :

(A) $\vec{a}.\vec{b} = 0,\ \vec{b}.\vec{c} = 0$ (B) $\vec{c}.\vec{a} = 0,\ \vec{a}.\vec{b} = 0$ (C) $\vec{a}.\vec{c} = 0,\ \vec{b}.\vec{c} = 0$ (D) $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$

Ans. D

Sol. $\sin\theta\cos\phi = 1 \Rightarrow \quad \theta = \dfrac{\pi}{2}$ and $\phi = 0 \quad \Rightarrow \quad \vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular

2. The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + 4\hat{k}$ are so placed that the end point of one vector is the starting point of the next vector. Then the vectors are :

(A) not coplanar
(B) coplanar but cannot form a triangle
(C) coplanar but can form a triangle
(D) coplanar and can form a right angled triangle

सदिशों $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ तथा $\vec{c} = 3\hat{i} + \hat{j} + 4\hat{k}$ इस प्रकार है कि एक सदिश का छोर अगले सदिश का प्रारम्भिक बिन्दु है, तो सदिश है :

(A) समतलीय नही है
(B) समतलीय है किन्तु त्रिभुज नही बनाते
(C) समतलीय है लेकिन त्रिभुज बनाते है
(D) समतलीय है तथा समकोण त्रिभुज बनाते है

Ans. B

Sol. Note that $\vec{a} + \vec{b} = \vec{c}$]

3. Given the vectors

$$\vec{u} = 2\hat{i} - \hat{j} - \hat{k}$$

$$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{w} = \hat{i} - \hat{k}$$

If the volume of the parallelopiped having – $c\vec{u}$, $\vec{v}$ and $c\vec{w}$ as concurrent edges, is 8 then 'c' can be equal to :

दिया गया है कि सदिश

$$\vec{u} = 2\hat{i} - \hat{j} - \hat{k}$$

$$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{w} = \hat{i} - \hat{k}$$

यदि समान्तरषट्फलक का आयतन जिसकी संगामी भुजाये – $c\vec{u}, \vec{v}$ तथा $c\vec{w}$ है, 8 है तो 'c' का मान होगा :

(A) ± 2 (B) 4 (C) 8 (D) 21

Ans. A

Sol. $V = -c^2[\vec{u}\,\vec{v}\,\vec{w}] = -c^2\begin{vmatrix} 2 & -1 & -1 \\ 1 & -1 & 2 \\ 1 & 0 & -1 \end{vmatrix} = -c^2[2(1-0)-1(1)+(-2-1)] = -c^2[2-1-3] = 8$

$\therefore \quad 2c^2 = 8 \quad \Rightarrow \quad c = 2 \text{ or } -2$ Ans.

4. Given $\bar{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\bar{b} = \hat{i} - \hat{j} + \hat{k}$, $\bar{c} = \hat{i} + 2\hat{j}$; $(\bar{a} \wedge \bar{b}) = \frac{\pi}{2}$, $\bar{a} \cdot \bar{c} = 4$ then :

दिया गया है $\bar{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\bar{b} = \hat{i} - \hat{j} + \hat{k}$, $\bar{c} = \hat{i} + 2\hat{j}$; $(\bar{a} \wedge \bar{b}) = \frac{\pi}{2}$, $\bar{a} \cdot \bar{c} = 4$ तो :

(A) $[\bar{a}\ \bar{b}\ \bar{c}]^2 = |\bar{a}|$ (B) $[\bar{a}\ \bar{b}\ \bar{c}] = |\bar{a}|$

(C) $[\bar{a}\ \bar{b}\ \bar{c}] = 0$ (D) $[\bar{a}\ \bar{b}\ \bar{c}] = |\bar{a}|^2$

Ans. D

Sol. $\left.\begin{matrix} x - y + 2 = 0 \quad \text{and} \\ x + 2y = 4 \end{matrix}\right] \quad \Rightarrow \quad x = 0\,;\, y = 2 \quad \Rightarrow \quad \vec{a} = 2\hat{j} + 2\hat{k} \ \ \& \ \ |\vec{a}|^2 = 8$

$[\bar{a}\ \bar{b}\ \bar{c}] = \begin{vmatrix} 0 & 2 & 2 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = 8 = |\bar{a}|^2 \Rightarrow$ (D)

5. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$ then $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 =$

माना $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ तीन अशून्य सदिश इस प्रकार है कि $\vec{c}$ एक इकाई सदिश है जो $\vec{a}$ तथा $\vec{b}$ दोनों के लम्बवत् है। यदि $\vec{a}$ तथा $\vec{b}$ में कोण $\frac{\pi}{6}$ है, तो $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 =$

(A) 0 (B) 1

(C) $\frac{1}{4}(a_1^2+a_2^2+a_3^2)(b_1^2+b_2^2+b_3^2)$ (D) $\frac{3}{4}(a_1^2+a_2^2+a_3^2)(b_1^2+b_2^2+b_3^2)(c_1^2+c_2^2+c_3^2)$

Ans. C

Sol. $(\vec{a}\times\vec{b}\cdot\vec{c})^2=|\vec{a}|^2\,|\vec{b}|^2|\vec{c}|^2\sin^2\theta\cos^2\phi=(a_1^2+a_2^2+a_3^2)(b_1^2+b_2^2+b_3^2)\frac{1}{4}\Rightarrow$ (C)

6. For three vectors $\vec{u},\vec{v},\vec{w}$ which of the following expressions is not equal to any of the remaining three?

तीन सदिशो $\vec{u},\vec{v},\vec{w}$ के लिये कौनसा व्यंजक अन्य तीन के बराबर नही है ?

(A) $\vec{u}.(\vec{v}\times\vec{w})$ (B) $(\vec{v}\times\vec{w}).\vec{u}$ (C) $\vec{v}.(\vec{u}\times\vec{w})$ (D) $(\vec{u}\times\vec{v}).\vec{w}$

Ans. C

Sol.

7. Let $\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=\hat{j}+\hat{k}$ and $\vec{c}=\alpha\vec{a}+\beta\vec{b}$. If the vectors, $\hat{i}-2\hat{j}+\hat{k}$, $3\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}$ are coplanar then $\frac{\alpha}{\beta}$ is :

माना $\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=\hat{j}+\hat{k}$ तथा $\vec{c}=\alpha\vec{a}+\beta\vec{b}$ है। यदि सदिश $\hat{i}-2\hat{j}+\hat{k}$, $3\hat{i}+2\hat{j}-\hat{k}$ तथा $\vec{c}$ समतलीय है तो $\frac{\alpha}{\beta}$ होगा :

(A) 1 (B) 2 (C) 3 (D) – 3

Ans. D

Sol. $\vec{V}_1=\hat{i}-2\hat{j}+\hat{k}$

$\vec{V}_2=3\hat{i}+2\hat{j}-\hat{k}$

$\vec{V}_3=\vec{c}=\alpha\vec{a}+\beta\vec{b}=\alpha(\hat{i}+\hat{j})+\beta(\hat{j}+\hat{k})=\alpha\hat{i}+(\alpha+\beta)\hat{j}+\beta\hat{k}=\vec{c}$

since $\vec{V}_1,\vec{V}_2,\vec{V}_3$ are coplanar

now $\begin{vmatrix}1&-2&1\\3&2&-1\\\alpha&\alpha+\beta&\beta\end{vmatrix}=0$, using $C_2\to C_2-(C_1+C_3)$, we get

$\begin{vmatrix}1&-4&1\\3&0&-1\\\alpha&0&\beta\end{vmatrix}=0$, hence $4(3\beta+\alpha)=0\Rightarrow 3\beta+\alpha=0\Rightarrow\frac{\alpha}{\beta}=-3$ Ans.

Aliter: $\vec{c} = \lambda(\vec{a} + \vec{b})$

$\vec{c}\,\vec{v} = \lambda(\vec{a} + \vec{b}) \cdot \vec{c} = 10 = \lambda(\vec{a}\,\vec{b}\,\vec{c}) = 10$]

8. A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ radians/sec. If $\vec{\omega}$ points in the direction of $\hat{i} + \hat{j} + \hat{k}$ then the equation to the locus of the points having tangential speed 20 m/sec. is :

एक दृढ़ पिण्ड मूल बिन्दु से गुजरने वाले अक्ष के सापेक्ष $10\sqrt{3}$ रेडियन/सेकण्ड के कोणीय वेग से घूर्णन कर रहा है। यदि $\vec{\omega}$ सदिश $\hat{i} + \hat{j} + \hat{k}$ की दिशा में है तो उन बिन्दुओं का बिन्दुपथ जिनका स्पर्श रेखीय वेग 20 m/sec है :

(A) $x^2 + y^2 + z^2 - xy - yz - zx - 1 = 0$ (B) $x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 1 = 0$

(C) $x^2 + y^2 + z^2 - xy - yz - zx - 2 = 0$ (D) $x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 2 = 0$

Ans. C

Sol. $\vec{n} = \dfrac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$; $\vec{\omega} = |\vec{\omega}|\,\vec{n} = 10\,(\hat{i} + \hat{j} + \hat{k})$

Now $\vec{v} = \vec{\omega} \times \vec{r} = 10\left(\hat{i} + \hat{j} + \hat{k}\right) \times \left(x\hat{i} + y\hat{j} + z\hat{k}\right)$ where $\vec{r}$ is the position vector of the point whose locus is to be determined.

Hence $\vec{v} = 10\left((z-y)\hat{i} - (z-x)\hat{j} + (y-x)\hat{k}\right)$

$\therefore\ |\vec{v}| = 10\sqrt{(x-y)^2 + (y-z)^2 + (z-x)^2}$ hence $2(x^2 + y^2 + z^2 - xy - yz - zx) = 4$

$\Rightarrow$ $x^2 + y^2 + z^2 - zy - yz - zx - 2 = 0$ which is the equation of a cylinder

9. Given 3 vectors $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$; $\vec{V}_2 = b\hat{i} + c\hat{j} + a\hat{k}$; $\vec{V}_3 = c\hat{i} + a\hat{j} + b\hat{k}$. In which one of the following conditions $\vec{V}_1$, $\vec{V}_2$ and $\vec{V}_3$ are linearly independent?

(A) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(D) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

दिया गया है 3 सदिश $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$; $\vec{V}_2 = b\hat{i} + c\hat{j} + a\hat{k}$; $\vec{V}_3 = c\hat{i} + a\hat{j} + b\hat{k}$ निम्न में से किन शर्तो के लिये $\vec{V}_1$, $\vec{V}_2$ तथा $\vec{V}_3$ रैखिक स्वतंत्र है?

(A) $a + b + c = 0$ तथा $a^2 + b^2 + c^2 \neq ab + bc + ca$

(B) $a + b + c = 0$ तथा $a^2 + b^2 + c^2 = ab + bc + ca$

(C) $a + b + c \neq 0$ तथा $a^2 + b^2 + c^2 = ab + bc + ca$

(D) $a + b + c \neq 0$ तथा $a^2 + b^2 + c^2 \neq ab + bc + ca$

Ans. D

Sol. only in (D) $[\vec{V}_1\ \vec{V}_2\ \vec{V}_3] \neq 0$

10. Given unit vectors $\vec{m}$, $\vec{n}$ and $\vec{p}$ such that angle between $\vec{m}$ and $\vec{n}$ = angle between $\vec{p}$ and $(\vec{m}\times\vec{n}) = \pi/6$ then $[\vec{n}\ \vec{p}\ \vec{m}] =$

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) None

दिया गया है कि इकाई सदिश $\vec{m}$, $\vec{n}$ तथा $\vec{p}$ इस प्रकार है कि $\vec{m}$ तथा $\vec{n}$ के मध्य कोण = $\vec{p}$ तथा $(\vec{m}\times\vec{n})$ के मध्य कोण = $\pi/6$ तो $[\vec{n}\ \vec{p}\ \vec{m}] =$

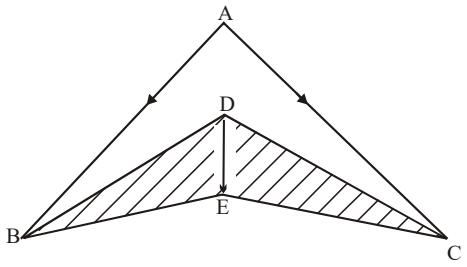
(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) कोई नहीं

Ans. A

Sol. $[\vec{n}\,\vec{p}\,\vec{m}] = \sin\theta\cos\phi = \sin\frac{\pi}{6}.\cos\frac{\pi}{6} = \frac{1}{2}.\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$

11. Let $\overrightarrow{AB} = 3\hat{i} - j$, $\overrightarrow{AC} = 2\hat{i} + 3\hat{j}$ and $\overrightarrow{DE} = 4\hat{i} - 2\hat{j}$. The area of the shaded region in the adjacent figure, is :

माना $\overrightarrow{AB} = 3\hat{i} - j$, $\overrightarrow{AC} = 2\hat{i} + 3\hat{j}$ तथा $\overrightarrow{DE} = 4\hat{i} - 2\hat{j}$ चित्र में रंगित किये गये क्षेत्र का क्षेत्रफल क्या होगा :



(A) 5 (B) 6 (C) 7 (D) 8

Ans. C

Sol. Given $\left.\begin{matrix}\vec{b} - \vec{a} = 3\hat{i} - \hat{j} \\ \vec{c} - \vec{a} = 2\hat{i} + 3\hat{j}\end{matrix}\right]$ and $\vec{e} - \vec{d} = 4\hat{i} - 2\hat{j}$

$$\text{Area} = \frac{1}{2}\left|\left(\overrightarrow{BE}\times\overrightarrow{BD}\right)+\left(\overrightarrow{EC}\times\overrightarrow{ED}\right)\right| = \frac{1}{2}\left[\left(\vec{e}-\vec{b}\right)\times\left(\vec{d}-\vec{b}\right)+\left(\vec{c}-\vec{e}\right)\times\left(\vec{d}-\vec{e}\right)\right]$$

$$= \frac{1}{2}\left[\vec{e}\times\vec{d}-\vec{e}\times\vec{b}-\vec{b}\times\vec{d}+\vec{c}\times\vec{d}-\vec{c}\times\vec{e}-\vec{e}\times\vec{d}\right]$$

$$= \frac{1}{2}\left[\left(\vec{e}\times\vec{c}-\vec{e}\times\vec{b}\right)-\vec{b}\times\vec{d}+\vec{c}\times\vec{d}\ \right] = \frac{1}{2}\left[\vec{e}\times\left(\vec{c}-\vec{b}\right)-\left(\vec{b}-\vec{c}\right)\times\vec{d}\right] = \frac{1}{2}\left[\left(\vec{b}-\vec{c}\right)\times\vec{e}-\left(\vec{b}-\vec{c}\right)\times\vec{d}\right]$$

$$= \frac{1}{2}\left[\left(\vec{b}-\vec{c}\right)\times\left(\vec{e}-\vec{d}\right)\right]$$

$$= \frac{1}{2}\left[\left\{\left(3\hat{i}-\hat{j}\right)-\left(2\hat{i}-3\hat{j}\right)\right\}\times\left(4\hat{i}-2\hat{j}\right)\right]$$ [13th, 13-12-2009, P-2]

$$= \frac{1}{2}\left[\left(\hat{i}-4\hat{j}\right)\times\left(4\hat{i}-2\hat{j}\right)\right] = 7 \text{ sq. units. Ans.}$$

12. A variable plane forms a tetrahedron of constant volume 64 K^3 with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is :

एक चर समतल निर्देशांक अक्षों एवं मूलबिन्दु के साथ एक नियत आयतन 64 K^3 का एक चतुष्फलक बनाता है, तो चतुष्फलक के केन्द्रक का बिन्दुपथ है :

(A) $x^3 + y^3 + z^3 = 6K^3$ (B) $xyz = 6k^3$ (C) $x^2 + y^2 + z^2 = 4K^2$ (D) $x^{-2} + y^{-2} + z^{-2} = 4K^{-2}$

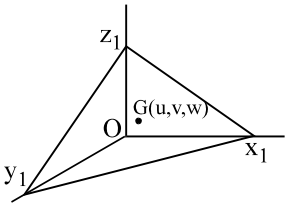
Ans. B

Sol. $\frac{x_1}{4} = u, \quad \frac{y_1}{4} = v, \quad \frac{z_1}{4} = w$

$x_1 = 4u, \qquad y_1 = 4v, z_1 = 4w$

$$V = \frac{1}{6}\begin{vmatrix} 4u & 0 & 0 \\ 0 & 4v & 0 \\ 0 & 0 & 4w \end{vmatrix} = \left(\frac{64}{6}\right)uvw$$

$$\therefore \qquad 64\cdot\left(\frac{uvw}{6}\right) = 64\ k^3 \quad \Rightarrow \quad xyz = 6k^3$$



13. Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be 3, 4 and 5 sq. units respectively. Then the area of the triangle BCD, is :

माना ABCD एक चतुष्फलक है जिसकी भुजायें AB, AC तथा AD परस्पर लम्बवत् है। माना त्रिभुजों ABC, ACD तथा ADB के क्षेत्रफल क्रमशः 3, 4 तथा 5 है तो त्रिभुज BCD का क्षेत्रफल :

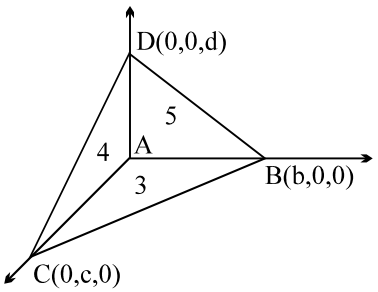
(A) $5\sqrt{2}$ (B) 5 (C) $\frac{5}{\sqrt{2}}$ (D) $\frac{5}{2}$

Ans. A

Sol. Area of $\Delta BCD = \frac{1}{2}\left|\overrightarrow{BC}\times\overrightarrow{BD}\right| = \frac{1}{2}\left|(b\hat{i}-c\hat{j})\times(b\hat{i}-d\hat{k})\right|$

$$= \frac{1}{2}\left|bd\,\hat{j}+bc\,\hat{k}+dc\,\hat{i}\right|$$

$$= \frac{1}{2}\sqrt{b^2c^2+c^2d^2+d^2b^2} \quad \text{_____(1)}$$



now $6 = bc$; $8 = cd$; $10 = bd$

$b^2c^2 + c^2d^2 + d^2b^2 = 200$

substituting the value in (1)

$A = \frac{1}{2}\sqrt{200} = 5\sqrt{2}$ Ans.

14. A parallelopiped is formed by planes drawn through the points (1, 2, 3) and (9, 8, 5) parallel to the coordinate planes then which of the following is not the length of an edge of this rectangular parallelopiped :

बिन्दुओं (1, 2, 3) तथा (9, 8, 5) से गुजरने वाले तथा निर्देशांक तलों के समानान्तर तलों द्वारा बनाये गये समान्तर षठफलक की निम्न में से कौनसी भुजा की लम्बाई नही होगी :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. $\left.\begin{array}{l} x=9\ ;\ x=1 \\ y=8\ ;\ y=2 \\ z=5\ ;\ z=3 \end{array}\right\} \Rightarrow$ edges of the cuboid are 8, 6, 2 $\Rightarrow$ (B) is correct

ONE OR MORE THAN ONE IS/ARE CORRECT

15. Vector $\vec{V}$ is perpendicular to the plane of vectors $\vec{a} = 2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{b} = \hat{i}-2\hat{j}+3\hat{k}$ and satisfies the condition $\vec{V}\cdot(\hat{i}+2\hat{j}-7\hat{k}) = 10$. Find $\left|\vec{V}\right|^2$.

सदिशो $\vec{a} = 2\hat{i}-3\hat{j}+\hat{k}$ तथा $\vec{b} = \hat{i}-2\hat{j}+3\hat{k}$ के तल के लम्बवत् सदिश $\vec{V}$ है तथा शर्त $\vec{V}\cdot(\hat{i}+2\hat{j}-7\hat{k}) = 10$ को संतुष्ट करता है। $\left|\vec{V}\right|^2$. ज्ञात कीजिए।

Ans. 75

Sol. $\vec{V} = \lambda(\vec{a} \times \vec{b}) = \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 1 \\ 1 & -2 & 3 \end{vmatrix}$

but $\vec{V} \cdot (\hat{i} + 2\hat{j} - 7\hat{k}) = 10$

$$\lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 1 \\ 1 & -2 & 3 \end{vmatrix} \cdot (\hat{i} + 2\hat{j} - 7\hat{k}) = 10$$

$$\lambda \begin{vmatrix} 1 & 2 & -7 \\ 2 & -3 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 10$$

$\lambda [(-9 + 2) - 2(6 - 1) - 7(-4 + 3)] = 10$

$\lambda [-7 - 10 + 7] = 10 \Rightarrow \lambda = -1$

Hence $\vec{V} = -(\vec{a} \times \vec{b}) = (\vec{b} \times \vec{a})$

$|\vec{V}|^2 = \vec{b}^2 \vec{a}^2 - (\vec{a} \cdot \vec{b})^2 = (14)(14) - (11)^2 = 196 - 121 = 75$ Ans.

16. Let two non-collinear vectors $\vec{a}$ and $\vec{b}$ inclined at an angle $\frac{2\pi}{3}$ be such that $|\vec{a}| = 3$ and $|\vec{b}| = 4$. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given as $\overrightarrow{OP} = (e^t + e^{-t})\vec{a} + (e^t - e^{-t})\vec{b}$. If the least distance of P from origin is $\sqrt{2}\sqrt{\sqrt{a} - b}$ where a, b ∈ N then find the value of (a + b).

माना दो अरेखीय सदिश $\vec{a}$ तथा $\vec{b}$ है जो परस्पर $\frac{2\pi}{3}$ पर झुके हुये है साथ ही $|\vec{a}| = 3$ एवं $|\vec{b}| = 4$ है। बिन्दु P इस प्रकार गमन कर रहा है कि स्थिति सदिश $\overrightarrow{OP}$ (जहाँ O मूल बिन्दु है) को $\overrightarrow{OP} = (e^t + e^{-t})\vec{a} + (e^t - e^{-t})\vec{b}$ से दर्शाया जाता है। यदि P की मूल बिन्दु से न्यूनतम दूरी $\sqrt{2}\sqrt{\sqrt{a} - b}$ है जहाँ a, b ∈ N है तो (a + b) का मान ज्ञात कीजिए।

Ans. 488

Sol. We have $\left(\overrightarrow{OP}\right)^2 = (e^t + e^{-t})^2 (\vec{a})^2 + (e^t - e^{-t})^2 (\vec{b})^2 + 2(e^t + e^{-t})(e^t - e^{-t})(\vec{a} \cdot \vec{b})$

$$\left((\vec{a})^2 = |\vec{a}|^2 = 9, (\vec{b})^2 = |\vec{b}|^2 = 16 \text{ and } \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\frac{2\pi}{3}\right)$$

$\Rightarrow \left|\overrightarrow{OP}\right|^2 = 9(e^t + e^{-t})^2 + 16(e^t - e^{-t})^2 + 2(e^{2t} - e^{-2t}) \cdot 3 \cdot 4 \cdot \left(-\frac{1}{2}\right) = 13e^{2t} + 37e^{-2t} - 14$

Now $\frac{d}{dt}\left|\overrightarrow{OP}\right|^2 = 0 \Rightarrow 26e^{2t} - 74e^{-2t} = 0 \Rightarrow e^{4t} = \frac{37}{13}$, so $e^{2t} = \frac{\sqrt{37}}{\sqrt{13}}$ [P Test-7, Vector]

$\therefore \quad \left|\overrightarrow{OP}\right|^2_{min} = 13\left(\frac{\sqrt{37}}{\sqrt{13}}\right) + 37\left(\frac{\sqrt{13}}{\sqrt{37}}\right) - 14 \Rightarrow \left|\overrightarrow{OP}\right|^2_{min} = 2\sqrt{481} - 14$

$\Rightarrow \quad \left|\overrightarrow{OP}\right|_{min} = \sqrt{2}\sqrt{\sqrt{481} - 7} = \sqrt{2}\sqrt{\sqrt{a} - b}$, so $a = 481, b = 7$

Hence $(a + b) = 488$. Ans.

1. The altitude of a parallelopiped whose three coterminous edges are the vectors, $\vec{A}=\hat{i}+\hat{j}+\hat{k}$; $\vec{B}=2\hat{i}+4\hat{j}-\hat{k}$ and $\vec{C}=\hat{i}+\hat{j}+3\hat{k}$ with $\vec{A}$ and $\vec{B}$ as the sides of the base of the parallelopiped, is :

(A) $\frac{2}{\sqrt{19}}$ (B) $\frac{4}{\sqrt{19}}$ (C) $\frac{2\sqrt{38}}{19}$ (D) None

एक समान्तर षट्फलक का लम्ब, जिसकी संगामी भुजायें सदिशों $\vec{A}=\hat{i}+\hat{j}+\hat{k}$; $\vec{B}=2\hat{i}+4\hat{j}-\hat{k}$ तथा $\vec{C}=\hat{i}+\hat{j}+3\hat{k}$ द्वारा दी जाती है। जिसमें $\vec{A}$ तथा $\vec{B}$ आधार की भुजाये है :

(A) $\frac{2}{\sqrt{19}}$ (B) $\frac{4}{\sqrt{19}}$ (C) $\frac{2\sqrt{38}}{19}$ (D) कोई नहीं

Ans. C

Sol. $h=\frac{2\left[\vec{A}\,\vec{B}\,\vec{C}\right]}{\left|\vec{A}\times\vec{B}\right|}$ and $\left|\vec{A}\times\vec{B}\right|=\sqrt{\vec{a}^2b^2-\left(\vec{a}.\vec{b}\right)^2}$

2. Consider ΔABC with $A\equiv(\bar{a})$; $B\equiv(\bar{b})$ and $C\equiv(\bar{c})$. If $\bar{b}\,.\,(\bar{a}+\bar{c})=\bar{b}\,.\,\bar{b}+\bar{a}\,.\,\bar{c}$; $|\bar{b}-\bar{a}|=3$; $|\bar{c}-\bar{b}|=4$ then the angle between the medians $\vec{AM}$ and $\vec{BD}$ is :

माना ΔABC जहाँ $A\equiv(\bar{a})$; $B\equiv(\bar{b})$ तथा $C\equiv(\bar{c})$ है। यदि $\bar{b}\,.\,(\bar{a}+\bar{c})=\bar{b}\,.\,\bar{b}+\bar{a}\,.\,\bar{c}$; $|\bar{b}-\bar{a}|=3$; $|\bar{c}-\bar{b}|=4$ तो माध्यिकाओं $\vec{AM}$ तथा $\vec{BD}$ के मध्य कोण होगा :

(A) $\pi-\cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (B) $\pi-\cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$ (C) $\cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (D) $\cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$

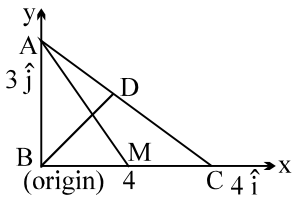
Ans. A

Sol. $\vec{b}\,.\,\vec{a}+\vec{b}\,.\,\vec{c}=\vec{b}\,.\,\vec{b}+\vec{a}\,.\,\vec{c}$

or $\vec{b}\,.\left(\vec{a}-\vec{b}\right)-\vec{c}\,.\left(\vec{a}-\vec{b}\right)=0$

or $\left(\vec{b}-\vec{c}\right).\left(\vec{a}-\vec{b}\right)=0$

$\Rightarrow$ BC and AB are perpendicular



Now find angle between AM & BM

3. If A (– 4, 0, 3) ; B (14, 2, –5) then which one of the following points lie on the bisector of the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$: ('O' is the origin of reference)

यदि A (– 4, 0, 3) ; B (14, 2, –5) तो निम्न में से कौनसा बिन्दु सदिश $\overrightarrow{OA}$ तथा $\overrightarrow{OB}$ के मध्य कोणार्द्धक पर स्थित होगा : ('O' मूल बिन्दु है)

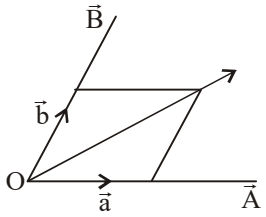
(A) (2, 1, –1) (B) (2, 11, 5) (C) (10, 2, –2) (D) (1, 1, 2)

Ans. D

Sol. $\overrightarrow{OA} = -4\hat{i} + 3\hat{k}$; $\overrightarrow{OB} = 14\hat{i} + 2\hat{j} - 5\hat{k}$

$\hat{a} = \frac{-4\hat{i} + 3\hat{k}}{5}$; $\hat{b} = \frac{14\hat{i} + 2\hat{j} - 5\hat{k}}{15}$

$\vec{r} = \frac{\lambda}{15}\left[-12\hat{i} + 9\hat{k} + 14\hat{i} + 2\hat{j} - 5\hat{k}\right]$



$\vec{r} = \frac{\lambda}{15}\left[2\hat{i} + 2\hat{j} + 4\hat{k}\right]$

$\vec{r} = \frac{2\lambda}{15}\left[\hat{i} + \hat{j} + 2\hat{k}\right]$ $\Rightarrow$ answer is (D) with $\lambda = \frac{15}{2}$

4. Position vectors of the four angular points of a tetrahedron ABCD are A(3, – 2, 1); B(3, 1, 5); C(4, 0, 3) and D(1, 0, 0). Acute angle between the plane faces ADC and ABC is :

चतुष्फलक ABCD के चार कोणीय बिन्दु A(3, – 2, 1); B(3, 1, 5); C(4, 0, 3) तथा D(1, 0, 0) है। फलको ADC तथा ABC के मध्य न्यूनकोण होना चाहिये :

(A) $\tan^{-1}\left(\frac{5}{2}\right)$ (B) $\cos^{-1}\left(\frac{2}{5}\right)$ (C) $\text{cosec}^{-1}\left(\frac{5}{2}\right)$ (D) $\cot^{-1}\left(\frac{3}{2}\right)$

Ans. A

Sol. $\vec{V}_1 = 0\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{V}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{V}_3 = -2\hat{i} + 2\hat{j} - \hat{k}$

$\vec{n}_1 = \vec{V}_1 \times \vec{V}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3 & 4 \\ 1 & 2 & 2 \end{vmatrix}$

$\vec{n}_1 = (-2)\hat{i} - (0-4)\hat{j} + (0-3)\hat{k}$

$\vec{n}_1 = -2\hat{i} + 4\hat{j} - 3\hat{k}$

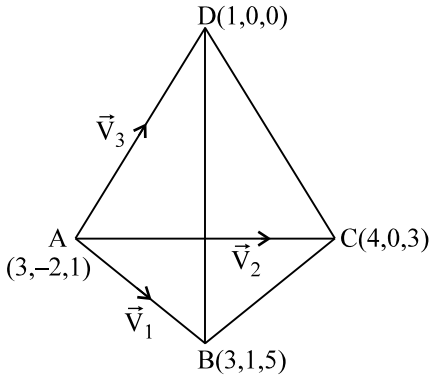
$\vec{n}_2 = \vec{V}_2 \times \vec{V}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ -2 & 2 & -1 \end{vmatrix}$

$\vec{n}_2 = (-2-4)\hat{i} - (-1+4)\hat{j} + (2+4)\hat{k}$

$\vec{n}_2 = -6\hat{i} - 3\hat{j} + 6\hat{k} = -3(2\hat{i} + \hat{j} - 2\hat{k})$

$\cos\theta = \dfrac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} = \dfrac{-4+4+6}{\sqrt{29}\cdot 3} = \dfrac{2}{\sqrt{29}}$

hence $\tan\theta = \dfrac{5}{2} \quad \Rightarrow \quad \theta = \tan^{-1}\left(\dfrac{5}{2}\right)$



5. The volume of the tetrahedron formed by the coterminus edges $\vec{a}$, $\vec{b}$, $\vec{c}$ is 3. Then the volume of the parallelepiped formed by the coterminus edges $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ is :

भुजाओं $\vec{a}$, $\vec{b}$, $\vec{c}$ द्वारा बनाये गये चतुष्फलक का आयतन 3 है। भुजाओं $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ द्वारा बनाये गये समान्तर षट् फलक का आयतन होगा :

(A) 6 (B) 18 (C) 36 (D) 9

Ans. C

Sol. $\dfrac{1}{6}[\bar{a}\,\bar{b}\,\bar{c}] = 3 \;\Rightarrow\; [\bar{a}\,\bar{b}\,\bar{c}] = 18$

volume of the required parallelepiped $= [\bar{a}+\bar{b}\;\; \bar{b}+\bar{c}\;\; \bar{c}+\bar{a}] = \{(\bar{a}+\bar{b})\times(\bar{b}+\bar{c})\}\cdot(\bar{c}+\bar{a})$

$= 2[\bar{a}\,\bar{b}\,\bar{c}] = 36$

6. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, then the vector $\vec{c}$ such that $\vec{a} \cdot \vec{c} = 2$ and $\vec{a} \times \vec{c} = \vec{b}$ is :

यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, तो सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} \cdot \vec{c} = 2$ तथा $\vec{a} \times \vec{c} = \vec{b}$ है, तो $\vec{c}$ होगा :

(A) $\frac{1}{3}\left(3\hat{i} - 2\hat{j} + 5\hat{k}\right)$ (B) $\frac{1}{3}\left(-\hat{i} + 2\hat{j} + 5\hat{k}\right)$ (C) $\frac{1}{3}\left(\hat{i} + 2\hat{j} - 5\hat{k}\right)$ (D) $\frac{1}{3}\left(3\hat{i} + 2\hat{j} + \hat{k}\right)$

Ans. B

Sol. $\vec{a} \times \vec{b} = \vec{a} \times (\vec{a} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} = 2\vec{a} - 3\vec{c}$

But $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix} = 3\hat{i} - 3\hat{k}$

Hence $3\vec{c} = 2\vec{a} - \left(3\hat{i} - 3\hat{k}\right) = \left(2\hat{i} + 2\hat{j} + 2\hat{k}\right) - \left(3\hat{i} - 3\hat{k}\right) = -\hat{i} + 2\hat{j} + 5\hat{k}$

$\Rightarrow \quad \vec{c} = \frac{1}{3}\left(-\hat{i} + 2\hat{j} + 5\hat{k}\right) \Rightarrow$ (B)

$\vec{a} \times \vec{c} = \vec{b}$
$\vec{a} \times (\vec{a} \times \vec{c}) = \vec{c} \times \vec{b}$
$(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} = 0$
$2\vec{a} - 3\vec{c} = \vec{c} \times \vec{b}$
$\vec{c} = \frac{2\vec{a} - \vec{a} \times \vec{b}}{3}$

7. $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, then the acute angle between $\vec{a}$ and $\vec{c}$ is :

$\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन सदिश जिनके परिमाण क्रमशः 1, 1 तथा 2 है। यदि $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, तो $\vec{a}$ तथा $\vec{c}$ के मध्य न्यूनकोण है:

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\frac{5\pi}{12}$

Ans. A

Sol. $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$; $|\vec{a}| = |\vec{b}| = 1$; $|\vec{c}| = 2$

$\vec{a} \times \vec{d} = -\vec{b} \quad \Rightarrow \quad (\vec{a} \times \vec{d})^2 = \vec{b}^2 = 1$

or $\quad |\vec{a}|^2 |\vec{d}|^2 - (\vec{a} \cdot \vec{d})^2 = 1$

or $\quad (\vec{a} \times \vec{c})^2 - 0 = 1 \Rightarrow |\vec{a}|^2 |\vec{c}|^2 - (\vec{a} \cdot \vec{c})^2 = 1$

$\Rightarrow \quad 4 - 2\cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{3}{4}; \quad \theta = \pi/6]$

[Alternative: $(\vec{a}.\vec{c})\vec{a} - (\vec{a}.\vec{a})\vec{c} = -\vec{b}$

$(\lambda\vec{a} - \vec{c})^2 = 1$ or $\lambda^2\vec{a}^2 + \vec{c}^2 - 2\lambda\vec{a}.\vec{c} = 1$ (where $\lambda = \vec{a} \cdot \vec{c}$)

$\Rightarrow \quad \lambda^2 + 4 - 2\lambda^2 = 1 \qquad \text{or} \qquad \lambda^2 = 3$

$\vec{a}^2\ \vec{c}^2 \cos^2\theta = 3$

$\cos^2\theta = 3/4$

$\theta = \pi/6$

8. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{3}$, then :

यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ तथा $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ रैखीय रूप से परतंत्र सदिश है तथा $|\vec{c}| = \sqrt{3}$ है, तो :

(A) $\alpha = 1, \beta = -1$ (B) $\alpha = 1, \beta = \pm 1$ (C) $\alpha = -1, \beta = \pm 1$ (D) $\alpha = \pm 1, \beta = 1$

Ans. D

Sol. $\alpha^2 + \beta^2 + 1 = 3 \quad \Rightarrow \quad \alpha^2 + \beta^2 = 2; \quad \text{and} \quad \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0$

9. A vector of magnitude $5\sqrt{5}$ coplanar with vectors $\hat{i} + 2\hat{j}$ and $\hat{j} + 2\hat{k}$ and the perpendicular vector $2\hat{i}+\hat{j}+2\hat{k}$ is :

एक सदिश जिसका परिमाण $5\sqrt{5}$ है जो सदिशों $\hat{i} + 2\hat{j}$ तथा $\hat{j} + 2\hat{k}$ के साथ समतलीय है तथा सदिश $2\hat{i}+\hat{j}+2\hat{k}$ से लंबवत् है :

(A) $\pm 5\left(5\hat{i} + 6\hat{j} - 8\hat{k}\right)$ (B) $\pm \sqrt{5}\left(5\hat{i} + 6\hat{j} - 8\hat{k}\right)$

(C) $\pm 5\sqrt{5}\left(5\hat{i} + 6\hat{j} - 8\hat{k}\right)$ (D) $\pm \left(5\hat{i} + 6\hat{j} - 8\hat{k}\right)$

Ans. D

Sol. Unit vector coplanar with $\vec{a}$ & $\vec{b}$ and perpendicular to vector $\vec{c}$ is $\pm \dfrac{(\vec{a} \times \vec{b}) \times \vec{c}}{|(\vec{a} \times \vec{b}) \times \vec{c}|}$

10. Let $\vec{\alpha} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{\beta} = \hat{i} + \hat{j}$. If $\vec{\gamma}$ is a unit vector, then the maximum value of $\left[\vec{\alpha} \times \vec{\beta} \quad \vec{\beta} \times \vec{\gamma} \quad \vec{\gamma} \times \vec{\alpha}\right]$ is equal to :

माना $\vec{\alpha} = 2\hat{i} + 3\hat{j} - \hat{k}$ तथा $\vec{\beta} = \hat{i} + \hat{j}$ यदि $\vec{\gamma}$ एक इकाई सदिश है, तो $\left[\vec{\alpha} \times \vec{\beta} \quad \vec{\beta} \times \vec{\gamma} \quad \vec{\gamma} \times \vec{\alpha}\right]$ का अधिकतम मान है :

(A) 2 (B) 3 (C) 4 (D) 9

Ans. B

Sol. As $\left[\vec{\alpha}\times\vec{\beta}\quad\vec{\beta}\times\vec{\gamma}\quad\vec{\gamma}\times\vec{\alpha}\right]=\left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]^2$

Now, $\left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]=\left(\vec{\alpha}\times\vec{\beta}\right)\cdot\vec{\gamma}=\left(\hat{i}-\hat{j}-\hat{k}\right)\cdot\vec{\gamma}$, which is maximum when angle between $\hat{i}-\hat{j}-\hat{k}$ and $\vec{\gamma}$ is zero and its maximum value $=\left|\hat{i}-\hat{j}-\hat{k}\right|\,|\vec{\gamma}|=\sqrt{3}$

$\therefore$ Maximum value of $\left[\vec{\alpha}\times\vec{\beta}\quad\vec{\beta}\times\vec{\gamma}\quad\vec{\gamma}\times\vec{\alpha}\right]=3$. Ans.

1. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar and $\vec{p},\vec{q},\vec{r}$ are reciprocal vectors to $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively, then $\left(\ell\vec{a}+m\vec{b}+n\vec{c}\right)\cdot\left(\ell\vec{p}+m\vec{q}+n\vec{r}\right)$ is equal to : (where *l*, m, n are scalars)

(A) $l^2+m^2+n^2$ (B) $l\,m+m\,n+n\,l$ (C) 0 (D) None of these

यदि $\vec{a},\vec{b},\vec{c}$ तीन विसमतलीय तथा $\vec{p},\vec{q},\vec{r}$ संदिशों $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ के प्रतिलोम (reciprocal) सदिश है, तो :

$\left(\ell\vec{a}+m\vec{b}+n\vec{c}\right)\cdot\left(\ell\vec{p}+m\vec{q}+n\vec{r}\right)$ = (जहाँ *l*, m, n अदिश है)

(A) $l^2+m^2+n^2$ (B) $l\,m+m\,n+n\,l$ (C) 0 (D) इनमें से कोई नहीं

Ans. A

Sol. $\vec{p}=\dfrac{\vec{b}\times\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$; $\vec{q}=\dfrac{\vec{c}\times\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$; $\vec{r}=\dfrac{\vec{a}\times\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$

Substitute the values of $\vec{p},\vec{q},\vec{r}$ to get the result

2. A plane P_1 has the equation $2x-y+z=4$ and the plane P_2 has the equation $x+ny+2z=11$. If the angle between P_1 and P_2 is $\dfrac{\pi}{3}$ then the value(s) of 'n' is/are :

समतल P_1 : $2x-y+z=4$ तथा समतल P_2 : $x+ny+2z=11$ है। यदि P_1 तथा P_2 के मध्य कोण $\dfrac{\pi}{3}$ है, तो 'n' के मान है :

(A) 7/2 (B) 17, –1 (C) –17, 1 (D) – 7/2

Ans. C

Sol. $\vec{n}_1=2\hat{i}-\hat{j}+\hat{k}$; $\vec{n}_2=\hat{i}+n\hat{j}+2\hat{k}$; $\cos\dfrac{\pi}{3}=\dfrac{2-n+2}{\sqrt{6}\sqrt{5+n^2}}=\dfrac{1}{2}$

3. The three vectors $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$, $\hat{k}+\hat{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to these three planes form a parallelopiped of volume :

तीन सदिश $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$, $\hat{k}+\hat{i}$ दो–दो बार में तीन समतलों का निर्माण करते है। इन समतलों के लम्बवत् इकाई सदिशों द्वारा बनाये गये समान्तर षट्फलक का आयतन है :

(A) $\frac{1}{3}$ (B) 4 (C) $\frac{3\sqrt{3}}{4}$ (D) $\frac{4}{3\sqrt{3}}$

Ans. D

Sol. $\left(\hat{i}+\hat{j}\right)\times\left(\hat{j}+\hat{k}\right)=\hat{i}-\hat{j}+\hat{k}\Rightarrow$ unit vector perpendicular as to the plane of

$\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is $\frac{1}{\sqrt{3}}\left(\hat{i}-\hat{j}+\hat{k}\right)$ similarly other two unit vectors are

$\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}-\hat{k}\right)$ and $\frac{1}{\sqrt{3}}\left(-\hat{i}+\hat{j}+\hat{k}\right)\Rightarrow V=\left[\hat{n}_1\ \hat{n}_2\ \hat{n}_3\right]=\frac{1}{3\sqrt{3}}\begin{vmatrix}1&-1&1\\1&1&-1\\-1&1&1\end{vmatrix}=\frac{4}{3\sqrt{3}}$

Alternatively: Let $\vec{a}=\hat{i}+\hat{j}$; $\vec{b}=\hat{j}+\hat{k}$ and $\vec{c}=\hat{k}+\hat{i}$.

Now $\left[\vec{a}\times\vec{b}\ ,\ \vec{b}\times\vec{c}\ ,\ \vec{c}\times\vec{a}\right]=\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2=\begin{vmatrix}1&1&0\\0&1&1\\1&0&1\end{vmatrix}^2=\left[1\,(1)-1\,(0-1)\right]^2=4$

Hence actual volume with unit vectors $=\frac{4}{\left|\vec{a}\times\vec{b}\right|\left|\vec{b}\times\vec{c}\right|\left|\vec{c}\times\vec{a}\right|}$

Now $\left|\vec{a}\times\vec{b}\right|=\sqrt{\vec{a}^2\ \vec{b}^2-(\vec{a}.\vec{b})^2}=\sqrt{4-1}=\sqrt{3}$ etc $V_{actual}=\frac{4}{3\sqrt{3}}$

4. If $\vec{x}$ and $\vec{y}$ are two non collinear vectors and a, b, c represent the sides of a ΔABC satisfying $(a-b)\vec{x}+(b-c)\vec{y}+(c-a)\left(\vec{x}\times\vec{y}\right)=0$ then ΔABC is :

(A) an acute angle triangle (B) an obtuse angle triangle
(C) a right angle triangle (D) a scalene triangle

यदि $\vec{x}$ तथा $\vec{y}$ दो विसमरेखीय सदिश है। a, b, c त्रिभुज ΔABC की भुजाऐं है जो $(a-b)\vec{x}+(b-c)\vec{y}+(c-a)\left(\vec{x}\times\vec{y}\right)=0$ तो ΔABC हैं :

(A) एक न्यूनकोणीय त्रिभुज (B) एक अधिकोणीय त्रिभुज
(C) एक समकोणीय त्रिभुज (D) एक विषमबाहु त्रिभुज

Ans. A

Sol. as $\vec{x}$, $\vec{y}$ and $\vec{x}\times\vec{y}$ are non coplanar vectors $\Rightarrow$ linearly independent
therefore $a-b=0=b-c=c-a\Rightarrow a=b=c\Rightarrow\Delta$ is equilateral $\Rightarrow$ A

5. Given three non – zero, non – coplanar vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ and $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$ if the vectors $\vec{r}_1 + 2\vec{r}_2$ and $2\vec{r}_1 + \vec{r}_2$ are collinear then (p, q) is :

दिया गया है कि तीन अशून्य, विसमतलीय सदिश $\vec{a}, \vec{b}, \vec{c}$ है तथा $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ एवं $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$ सदिश इसप्रकार है कि $\vec{r}_1 + 2\vec{r}_2$ तथा $2\vec{r}_1 + \vec{r}_2$ समरेखीय है, तो (p, q) होगा :

(A) (0, 0) (B) (1, –1) (C) (–1, 1) (D) (1, 1)

Ans. D

Sol. $\vec{r}_1 + 2\vec{r}_2 = (p\vec{a} + q\vec{b} + \vec{c}) + 2(\vec{a} + p\vec{b} + q\vec{c})$

$\vec{r}_1 + 2\vec{r}_2 = (p+2)\vec{a} + (q+2p)\vec{b} + (1+2q)\vec{c}$

$2\vec{r}_1 + \vec{r}_2 = (2p+1)\vec{a} + (2q+p)\vec{b} + (2+q)\vec{c}$

$$\frac{p+2}{2p+1} = \frac{q+2p}{2q+p} = \frac{1+2q}{2+q} = \frac{p+q+2p+2q+3}{p+q+2p+2q+3} = 1 \Rightarrow p = 1 \text{ and } q = 1$$

6. If the vectors $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar and l, m, n are distinct scalars, then

$\left[\left(l\vec{a} + m\vec{b} + n\vec{c}\right)\left(l\vec{b} + m\vec{c} + n\vec{a}\right)\left(l\vec{c} + m\vec{a} + n\vec{b}\right)\right] = 0$ implies :

यदि सदिश $\vec{a}, \vec{b}, \vec{c}$ विसमतलीय है तथा l, m, n तीन भिन्न अदिश है, तो

$\left[\left(l\vec{a} + m\vec{b} + n\vec{c}\right)\left(l\vec{b} + m\vec{c} + n\vec{a}\right)\left(l\vec{c} + m\vec{a} + n\vec{b}\right)\right] = 0$ का तात्पर्य है :

(A) $lm + mn + nl = 0$ (B) $l + m + n = 0$ (C) $l^2 + m^2 + n^2 = 0$ (D) $l^3 + m^3 + n^3 = 0$

Ans. B

Sol. $\left.\begin{aligned}\vec{V}_1 &= l\vec{a} + m\vec{b} + n\vec{c}\\ \vec{V}_2 &= n\vec{a} + l\vec{b} + m\vec{c}\\ \vec{V}_3 &= m\vec{a} + n\vec{b} + l\vec{c}\end{aligned}\right\}$ when $\vec{a}, \vec{b}, \vec{c}$ are non – coplanar

$$\therefore [\vec{V}_1\ \vec{V}_2\ \vec{V}_3] = \begin{vmatrix} l & m & n \\ n & l & m \\ m & n & l \end{vmatrix} [\vec{a}\,\vec{b}\,\vec{c}] = 0; \qquad \text{But } [\vec{a}\,\vec{b}\,\vec{c}] \neq 0 \Rightarrow \begin{vmatrix} l & m & n \\ n & l & m \\ m & n & l \end{vmatrix} = 0 \ \forall\ \lambda \in R$$

or $(l + m + n)[(l-m)^2 + (m-n)^2 + (n-l)^2] = 0 \Rightarrow l + m + n = 0$

7. The orthogonal projection A' of the point A with position vector (1, 2, 3) on the plane $3x - y + 4z = 0$ is :

बिन्दु A का स्थिति सदिश (1, 2, 3) है, का तल $3x - y + 4z = 0$ के सापेक्ष लम्बवत् प्रक्षेप है :

(A) $(-1, 3, -1)$ (B) $\left(-\frac{1}{2}, \frac{5}{2}, 1\right)$ (C) $\left(\frac{1}{2}, -\frac{5}{2}, -1\right)$ (D) $(6, -7, -5)$

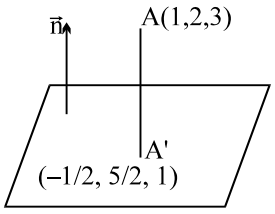
Ans. B

Sol. $\vec{n} = 3\hat{i} - \hat{j} + 4\hat{k}$

line through A and parallel to $\vec{n}$ is

$$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(3\hat{i} - \hat{j} + 4\hat{k})$$

$$= 3\lambda + 1, 2 - \lambda, 3 + 4\lambda \quad \text{_____(1)}$$



Hence (1) must satisfy the plane $3x - y + 4z = 0$

$$3(3\lambda + 1) - (2 - \lambda) + 4(3 + 4\lambda) = 0$$

$$26\lambda + 13 = 0 \Rightarrow \lambda = -\frac{1}{2}$$

Hence A' is $\left(-\frac{1}{2}, \frac{5}{2}, 1\right)$ which is the foot of the perpendicular from A on the given plane

8. If $\hat{a}$ and $\hat{b}$ are unit vectors then the vector defined as $\vec{V} = (\hat{a} \times \hat{b}) \times (\hat{a} + \hat{b})$ is collinear to the vector :

यदि $\hat{a}$ तथा $\hat{b}$ इकाई सदिश है, तो सदिश $\vec{V} = (\hat{a} \times \hat{b}) \times (\hat{a} + \hat{b})$ किसके साथ एक समरेखीय सदिश है :

(A) $\hat{a} + \hat{b}$ (B) $\hat{b} - \hat{a}$ (C) $2\hat{a} - \hat{b}$ (D) $\hat{a} + 2\hat{b}$

Ans. B

Sol. $\vec{V} = (\hat{a} \times \hat{b}) \times \hat{a} + (\hat{a} \times \hat{b}) \times \hat{b} = \hat{b} - (\hat{a} \cdot \hat{b})\hat{a} + (\hat{a} \cdot \hat{b})\hat{b} - \hat{a} = (\hat{b} - \hat{a}) + (\hat{b} - \hat{a})(\hat{a} \cdot \hat{b})$

$\vec{V} = (\hat{b} - \hat{a})(1 + \hat{a} \cdot \hat{b}) = \lambda(\hat{b} - \hat{a})$ Ans.

9. If $\hat{a}$ and $\hat{b}$ are orthogonal unit vectors then for any non zero vector $\vec{r}$, the vector $(\vec{r} \times \hat{a})$ equals :

यदि $\hat{a}$ तथा $\hat{b}$ लम्बवत् इकाई सदिश है, तो किसी अशून्य सदिश $\vec{r}$ के लिए $(\vec{r} \times \hat{a})$ बराबर है :

(A) $[\vec{r}\ \hat{a}\ \hat{b}]\ (\hat{a} + \hat{b})$ (B) $[\vec{r}\ \hat{a}\ \hat{b}]\hat{a} + (\vec{r} \cdot \hat{a})(\hat{a} \times \hat{b})$

(C) $[\vec{r}\ \hat{a}\ \hat{b}]\hat{b} + (\vec{r} \cdot \hat{b})(\hat{b} \times \hat{a})$ (D) $[\vec{r}\ \hat{a}\ \hat{b}]\hat{b} + (\vec{r} \cdot \hat{a})(\hat{a} \times \hat{b})$

Ans. C

Sol. Since $\hat{a}$ and $\hat{b}$ are perpendicular hence $\hat{a}$, $\hat{b}$ and $\hat{a} \times \hat{b}$ are non coplanar.

Hence any vector say $(\vec{r} \times \hat{a})$ can be expressed as

$\therefore \quad \vec{r}\times\hat{a} = x\hat{a} + y\hat{b} + z(\hat{a}\times\hat{b})$ _____(1)

dot with $\vec{a}$ $\quad 0 = x + 0 + 0 \Rightarrow x = 0$

dot with $\vec{b}$ $\quad [\vec{r}\ \hat{a}\ \hat{b}] = 0 + y + 0 \Rightarrow y = [\vec{r}\ \hat{a}\ \hat{b}]$

dot with $\vec{r}$ $\quad 0 = \underbrace{x\hat{a}\cdot\vec{r}}_{\text{zero as } x=0} + y\vec{r}\cdot\hat{b} + z[\vec{r}\,\hat{a}\,\hat{b}]$

$0 = [\vec{r}\ \hat{a}\ \hat{b}](\vec{r}\cdot\hat{b}) + z[\vec{r}\ \hat{a}\ \hat{b}]$

$\Rightarrow \quad z = -(\vec{r}\cdot\hat{b})$

Hence $\vec{r}\times\vec{a} = [\vec{r}\ \hat{a}\ \hat{b}]\hat{b} - (\vec{r}\cdot\hat{b})\ (\hat{a}\times\hat{b})$

$\vec{r}\times\hat{a} = [\vec{r}\ \hat{a}\ \hat{b}]\hat{b} + (\vec{r}\cdot\hat{b})\ (\hat{b}\times\hat{a})$

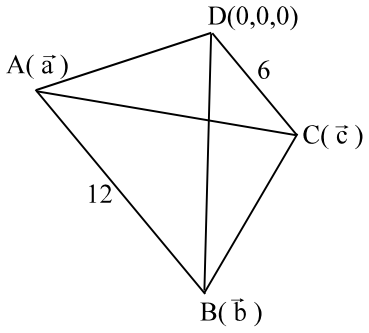
SUBJECTIVE TYPE

10. Given a tetrahedron D-ABC with AB = 12 , CD = 6. If the shortest distance between the skew lines AB and CD is 8 and the angle between them is $\frac{\pi}{6}$, then find the volume of tetrahedron.

दिया है कि एक चतुष्फलक D-ABC जहाँ AB = 12 , CD = 6 है। विसमतलीय रेखाओं AB तथा CD के मध्य न्यूनतम दूरी 8 है तथा उनके मध्य कोण $\frac{\pi}{6}$ है, तो चतुष्फलक का आयतन ज्ञात कीजिए।

Ans. 48

Sol. Given $|\vec{b} - \vec{a}| = 12$; $|\vec{c}| = 6$



Equation of CD is $\vec{r} = \lambda\vec{c}$

and eq. of AB is $\vec{r} = \vec{a} + \mu(\vec{b} - \vec{a})$

$$\text{S.D.} = \left|\frac{\vec{a}\cdot\vec{c}\times(\vec{b}-\vec{a})}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8$$

$$\left|\frac{[\vec{a}\ \ \vec{c}\ \ \vec{b}]}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8$$

$$\left|[\vec{a}\ \ \vec{c}\ \ \vec{b}]\right| = 8\left|\vec{c}\times(\vec{b}-\vec{a})\right|$$

$$6V = \frac{8|\vec{c}||\vec{b}-\vec{a}|}{2} \quad \Rightarrow \quad 6V = (4)(6)(12) \quad \Rightarrow \quad V = 48 \text{ Ans.}$$

11. Given $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{C} = \hat{i} + 2\hat{j} + \hat{k}$. Compute the value of $\left|\vec{A} \times \left(\vec{A} \times (\vec{A} \times \vec{B})\right) \cdot \vec{C}\right|$.

दिया है कि $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$ तथा $\vec{C} = \hat{i} + 2\hat{j} + \hat{k}$ तो $\left|\vec{A} \times \left(\vec{A} \times (\vec{A} \times \vec{B})\right) \cdot \vec{C}\right|$ का मान ज्ञात कीजिए।

Ans. 343

Sol. $\vec{V} = \vec{A} \times \left((\vec{A} \cdot \vec{B})\vec{A} - (\vec{A} \cdot \vec{A})\vec{B}\right) \cdot \vec{C}$

$$= \left(\underbrace{\vec{A} \times (\vec{A} \cdot \vec{B})\vec{A}}_{\text{zero}} - (\vec{A} \cdot \vec{A})\vec{A} \times \vec{B} \right) \cdot \vec{C}$$

$$= - |\vec{A}|^2 [\vec{A} \quad \vec{B} \quad \vec{C}]$$

now $|\vec{A}|^2 = 4 + 9 + 36 = 49$

$$[\vec{A} \quad \vec{B} \quad \vec{C}] = \begin{vmatrix} 2 & 3 & 6 \\ 1 & 1 & -2 \\ 1 & 2 & 1 \end{vmatrix} = 2(1 + 4) - 1(3 - 12) + 1(-6 - 6)$$

$$= 10 + 9 - 12 = 7$$

$\therefore \quad \left| - |\vec{A}|^2 [\vec{A}\vec{B}\vec{C}] \right| = 49 \times 7 = 343$ Ans.

12. A vector $\vec{V} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ satisfies the following conditions :

(i) magnitude of $\vec{V}$ is $7\sqrt{2}$

(ii) $\vec{V}$ is parallel to the plane $x - 2y + z = 6$

(iii) $\vec{V}$ is orthogonal to the vector $2\hat{i} - 3\hat{j} + 6\hat{k}$

and (iv) $\vec{V} \cdot \hat{i} > 0$.

Find the value of $(v_1 + v_2 + v_3)$.

एक सदिश $\vec{V} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ निम्न शर्तों को संतुष्ट करता है :

(i) $\vec{V}$ का परिमाप है $7\sqrt{2}$

(ii) $\vec{V}$ समतल $x - 2y + z = 6$ के समानान्तर है

(iii) $\vec{V}$ सदिश $2\hat{i} - 3\hat{j} + 6\hat{k}$ के लम्बवत् है

एवं (iv) $\vec{V} \cdot \hat{i} > 0$.

$(v_1 + v_2 + v_3)$ का मान ज्ञात कीजिए।

Ans. 12

Sol. A vector parallel to the plane $x-2y+z=6$ and orthogonal to $2\hat{i}-3\hat{j}+6\hat{k}$ is $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 2 & -3 & 6 \end{vmatrix}$

$\bar{x} = \hat{i}(-12+3) - \hat{j}(6-2) + \hat{k}(-3+4) = -9\hat{i} - 4\hat{j} + \hat{k}$

Unit vector $\hat{x} = \dfrac{(-9\hat{i}-4\hat{j}+\hat{k})}{\sqrt{81+16+1}} = \left(\dfrac{-9\hat{i}-4\hat{j}+\hat{k}}{7\sqrt{2}}\right)$

vector of magnitude $7\sqrt{2} = \pm(-9\hat{i}-4\hat{j}+\hat{k})$

hence $\vec{V}\cdot\hat{i} > 0 \Rightarrow$ coefficient of $\hat{i} > 0$, hence taking (–ve) sign we get

$\vec{V} = 9\hat{i}+4\hat{j}-\hat{k}$

$\therefore \Rightarrow v_1 + v_2 + v_3 = 12$ Ans.

13. Let $(\vec{p}\times\vec{q})\times\vec{r} + (\vec{q}\cdot\vec{r})\vec{q} = (x^2+y^2)\vec{q} + (14-4x-6y)\vec{p}$ and $(\vec{r}\cdot\vec{r})\vec{p} = \vec{r}$ where $\vec{p}$ and $\vec{q}$ are two non-zero non-collinear vectors and x and y are scalars. Find the value of $(x+y)$.

माना $(\vec{p}\times\vec{q})\times\vec{r} + (\vec{q}\cdot\vec{r})\vec{q} = (x^2+y^2)\vec{q} + (14-4x-6y)\vec{p}$ तथा $(\vec{r}\cdot\vec{r})\vec{p} = \vec{r}$ जहाँ $\vec{p}$ तथा $\vec{q}$ दो अशून्य विसमरेखीय सदिश है तथा x एवं y अदिश संख्याऐं है। $(x+y)$ का मान ज्ञात कीजिए।

Ans. 5

Sol. $(\vec{p}\cdot\vec{r})\vec{q} - (\vec{q}\cdot\vec{r})\vec{p} + (\vec{q}\cdot\vec{r})\vec{q} = (x^2+y^2)\vec{q} + (14-4x-6y)\vec{p}$

$\therefore \vec{p}\cdot\vec{r} + \vec{q}\cdot\vec{r} = x^2+y^2$ _____(1)

and $-(\vec{q}\cdot\vec{r}) = 14-4x-6y$ _____(2)

From (1) + (2)

$\vec{p}\cdot\vec{r} = x^2+y^2-4x-6y+14$ _____(3)

$\because (\vec{r}\cdot\vec{r})\vec{p} = \vec{r}$

Taking dot product with $\vec{r}$, we get

$(\vec{r}\cdot\vec{r})(\vec{p}\cdot\vec{r}) = \vec{r}\cdot\vec{r} \Rightarrow \vec{p}\cdot\vec{r} = 1$

$\therefore$ from(3)

$x^2+y^2-4x-6y+14 = 1$

$\Rightarrow (x-2)^2 + (y-3)^2 = 0 \Rightarrow x = 2$ and $y = 3$

Hence $(x+y) = 5$. Ans.